1. Protein of Interest

Name: PIEZO 1

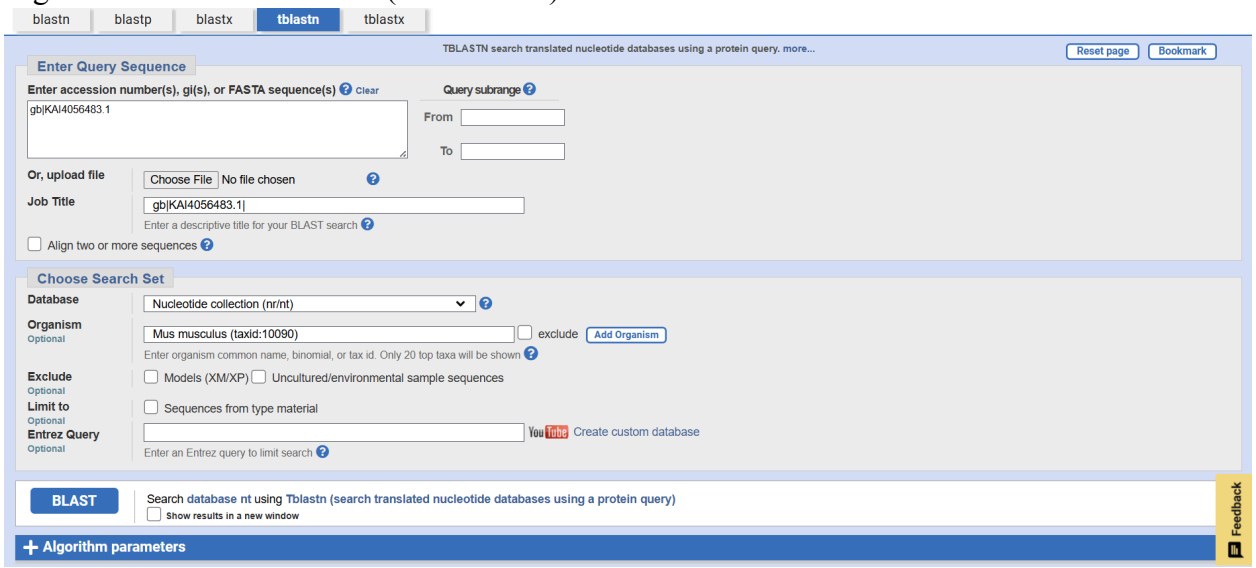
Accession: KAI4056483.**1**

Species: Homo Sapiens

Known function: PIEZO 1 protein is a mechanosensitive ion channel protein that allows calcium ions to transfer for muscle contraction in hearts.

2. BLAST Search

Method: TBLASTN search against nt
Database: Nucleotide collection (nt)
Organism: Mus musculus (taxid:10090)



‹ Edit Search | Save Search | Search Summary ⌄ | How to read this report? | BLAST Help Videos | Back to Traditional Results Page

Your search is limited to records **include: Mus musculus (taxid:10090)**

Job Title	gb|KAI4056483.1|
RID	FX7F5V68014 *Search expires on 10-28 06:17 am* Download All ⌄
Program	TBLASTN Citation ⌄
Database	nt See details ⌄
Query ID	KAI4056483.1
Description	piezo type mechanosensitive ion channel component 1 [H ...
Molecule type	amino acid
Query Length	2521
Other reports	

Filter Results

Organism *only top 20 will appear* exclude

Type common name, binomial, taxid or group name

+ Add organism

Percent Identity to | E value to | Query Coverage to

Filter | Reset

Descriptions | Graphic Summary | Alignments | Taxonomy

Sequences producing significant alignments Download ⌄ Select columns ⌄ Show 100

select all *80 sequences selected* GenBank Graphics

Description	Scientific Name	Max Score	Total Score	Query Cover	E value	Per. Ident	Acc. Len	Accession
Mus musculus piezo-type mechanosensitive ion channel component 1 (Piezo1), transcript variant 2, mRNA	Mus musculus	3488	3488	100%	0.0	70.48%	8216	NM_001037298.1
Mus musculus piezo-type mechanosensitive ion channel component 1 (Piezo1), transcript variant 1, mRNA	Mus musculus	3487	3487	100%	0.0	70.41%	8219	NM_001357349.1

Chosen match: Accession NM_001037298.1, a 8216 base pair Mus musculus mRNA. See below for more alignment details.

This alignment has an e-value of 0.0, 100% coverage, and 70.48% identity.

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Mus musculus piezo-type mechanosensitive ion channel component 1 (Piezo1), transcript variant 2, mRNA

Sequence ID: NM_001037298.1 Length: 8216 Number of Matches: 1

Range 1: 292 to 7929 GenBank Graphics ▼Next Match ▲Previous Match

Score	Expect	Method	Identities	Positives	Gaps	Frame
3488 bits(9044)	0.0	Compositional matrix adjust.	2054/2564(80%)	2215/2564(86%)	61/2564(2%)	+1

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Query  1     MEPHVLGAVLYWlllpcallaacllRFSGlslvyllfllllPWFPGPTRCGLQGHTGRll  60
             MEPHVLGA LYWLLLPC LLAA LLRF+ LSLVYLLFLLLLPW PGP+R  + GHTGRLL
Sbjct  292   MEPHVLGAGLYWLLLPCTLLAASLLRFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRLL  471

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             RALL LSLLFLVAHLA QICLHTVP LDQ LG + S W  +S+HIGVTRLDLKDI N  R
Sbjct  472   RALLCLSLLFLVAHLAFQICLHTVPHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTR  651

Query  121   LVAPDLGILVVSSVCLGICGRLARNTRQSPHPRELDDDERDVD------ASPTAGLQEAA  174
             LVAPDLG+L+ SS+CLG+CGRL R  RQS   +ELDDD+ D D      A+P  GL+ A
Sbjct  652   LVAPDLGVLLASSLCLGLCGRLTRKARQSRRTQELDDDDDDDDDDEDIDAAPAVGLKGAP  831

Query  175   TLAPTRRSRLAARFRVTAHWLLVAAGRVLAVTLLALAGIAHPsalssvylllflalCTWW  234
              LA  RR  LA+RFRVTAHWLL+ +GR L + LLALAGIAHPSA SSVYL++FLA+CTWW
Sbjct  832   ALATKRRLWLASRFRVTAHWLLMTSGRTLVIVLLALAGIAHPSAFSSVYLVVFLAICTWW  1011

Query  235   ACHFPISTRGFSRLCAAVGCFGAGHLICLYCYQMPLAQALLPPAGIWARVLGLKDFVGPT  294
             +CHFP+S  GF+ LC  V CFGAGHLICLYCYQ P  Q +LPP  IWAR+ GLK+FV
Sbjct  1012  SCHFPLSPLGFNTLCVMVSCFGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDLP  1191

Query  295   NCSSPHALVLNTGLDWPVYASPGVLLLLCYATASLRKLRAYRPSGQRKEAAKGYEARELE  354
             N SSP+ALVLNT   WP+Y SPG+LLLL Y   SL KL    PS  RKE  +  E  ELE
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Query  355   LAELDQWPQERESDQHVVPTAPDTEADNCIVHELTGQSSLLRRPVRPKRAEPGEASPLHS  414
             L  L+  PQ R++ Q  +P   + + DNC VH LT QS + +RPVRP+ AE  E SPLH
Sbjct  1372  LDHLEPEPQARDATQGEMPMTTEPDLDNCTVHVLTSQSPVRQRPVRPRLAELKEMSPLHG  1551

Query  415   LGHLIMDQSYVCALIAMMVWSITYHSWLTFVLLLWACLIWTVRSRHQLAMLCSPCILLYG  474
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Query  535   KEKLLKWAESPAALTEVTVADTEPTRTQTLLQSLGELVKGVYAKYWIYVCAGMFIVVSFA  594
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Related Information

Gene - associated gene details

GEO Profiles - microarray expression data

Genome Data Viewer - aligned genomic context

```
>Mus musculus piezo-type mechanosensitive ion channel component 1 (Piezo1),
transcript variant 2, mRNA
      Sequence ID: NM_001037298.1 Length: 8216
```

Range 1: 292 to 7929

Score:3488 bits(9044), Expect:0.0,
Method:Compositional matrix adjust.,
Identities:2054/2564(80%), Positives:2215/2564(86%), Gaps:61/2564(2%)

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Query  1     MEPHVLGAVLYWlllpcallaacllRFSGlslvyllfllllPWFPGPTRCGLQGHTGRll  60
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             RALL LSLLFLVAHLA QICLHTVP LDQ LG + S W  +S+HIGVTRLDLKDI N  R
Sbjct  472   RALLCLSLLFLVAHLAFQICLHTVPHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTR  651

Query  121   LVAPDLGILVVSSVCLGICGRLARNTRQSPHPRELDDDERDVD------ASPTAGLQEAA  174
             LVAPDLG+L+ SS+CLG+CGRL R  RQS   +ELDDD+ D D      A+P  GL+ A
Sbjct  652   LVAPDLGVLLASSLCLGLCGRLTRKARQSRRTQELDDDDDDDDDDEDIDAAPAVGLKGAP  831

Query  175   TLAPTRRSRLAARFRVTAHWLLVAAGRVLAVTLLALAGIAHPsalssvylllflalCTWW  234
              LA  RR  LA+RFRVTAHWLL+ +GR L + LLALAGIAHPSA SSVYL++FLA+CTWW
Sbjct  832   ALATKRRLWLASRFRVTAHWLLMTSGRTLVIVLLALAGIAHPSAFSSVYLVVFLAICTWW  1011

Query  235   ACHFPISTRGFSRLCAAVGCFGAGHLICLYCYQMPLAQALLPPAGIWARVLGLKDFVGPT  294
             +CHFP+S  GF+ LC  V CFGAGHLICLYCYQ P  Q +LPP  IWAR+ GLK+FV
Sbjct  1012  SCHFPLSPLGFNTLCVMVSCFGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDLP  1191

Query  295   NCSSPHALVLNTGLDWPVYASPGVLLLLCYATASLRKLRAYRPSGQRKEAAKGYEARELE  354
             N SSP+ALVLNT   WP+Y SPG+LLLL Y   SL KL    PS  RKE  +  E  ELE
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Query  355   LAELDQWPQERESDQHVVPTAPDTEADNCIVHELTGQSSLLRRPVRPKRAEPGEASPLHS  414
             L  L+  PQ R++ Q  +P   + + DNC VH LT QS + +RPVRP+ AE  E SPLH
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Query  595   GRLVVYKIVYMFLFLLCLTLFQVYYSLWRKLLKAFWWLVVAYTMLVLIAVYTFQFQDFPA  654
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Sbjct  2089  GRLVVYKIVYMFLFLLCLTLFQVYYTLWRKLLRVFWWLVVAYTMLVLIAVYTFQFQDFPT  2268

Query  655   YWRNLTGFTDEQLGDLGLEQFSVSELFSSILVPGFFLLACILQLHYFHRPFMQLTDMEHV  714
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Sbjct  2269  YWRNLTGFTDEQLGDLGLEQFSVSELFSSILIPGFFLLACILQLHYFHRPFMQLTDLEHV  2448

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               PGTR PRWAHRQDAVS  PLL  +++E               R++G  +  PHQATQV
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Sbjct  2593  PEGTASKWGLVADRLLDLAASFSAVLTRIQVFVRRLLELHVFKLVALYTVWVALKEVSVM  2772

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Sbjct  2773  NLLLVVLWAFALPYPRFRPMASCLSTVWTCIIIVCKMLYQLKIVNPHEYSSNCTEPFPNN  2952

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Sbjct  3133  RQHQQAPLPAQAVCADGTRQRLDQDLLSCLKYFINFFFYKFGLEICFLMAVNVIGQRMNF  3312

Query  1014  LVTLHGCWLVAILTRRHRQAIARLWPNyclflalfllyqyllclGMPPALCIDYPWRWSR  1073
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Sbjct  3313  MVILHGCWLVAILTRRRREAIARLWPNYCLFLTLFLLYQYLLCLGMPPALCIDYPWRWSK  3492

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              LEPLRGEPNP+PNFIHCRSYLDMLKVAVFRYLFWLVLVVVFV GATRISIFGLGYLLAC
Sbjct  3673  HLEPLRGEPNPIPNFIHCRSYLDMLKVAVFRYLFWLVLVVVFVAGATRISIFGLGYLLAC  3852

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             FYLLLFGT LLQ+DTRA+LVLWDCLILYNVTVIISKNMLSLL+CVFVEQMQ+ FCWVIQL
Sbjct  3853  FYLLLFGTTLLQKDTRAQLVLWDCLILYNVTVIISKNMLSLLSCVFVEQMQSNFCWVIQL  4032

Query  1254  FSLVCTVKGYYDPKEMMDRDQDCLLPVEEAGIIWDSVCfffllqrrvflSHYYLHVRAD  1313
             FSLVCTVKGYYDPKEMM RD+DCLLPVEEAGIIWDS+CFFFLLLQRR+FLSHY+LHV AD
Sbjct  4033  FSLVCTVKGYYDPKEMMTRDRDCLLPVEEAGIIWDSICFFFLLLQRRIFLSHYFLHVSAD  4212

Query  1314  LQATALLASRGFALYNAANLKSIDFHRRIEEKSLAQLKRQMERIRAKQEKHRQGRVDRSR  1373
             L+ATAL ASRGFALYNAANLKSI+FHR+IEEKSLAQLKRQM+RIRAKQEK+RQ +  R +
Sbjct  4213  LKATALQASRGFALYNAANLKSINFHRQIEEKSLAQLKRQMKRIRAKQEKYRQSQASRGQ  4392

Query  1374  PQDTLGPKDpglepgpdspggssppRRQWWRPWLDHATVIHSGDYFLFesdseeeeeavp  1433
              Q    P+DP  EPGPDSPGGSSPPRRQWWRPWLDHATVIHSGDYFLFESDSEEEEEA+P
Sbjct  4393  LQSK-DPQDPSQEPGPDSPGGSSPPRRQWWRPWLDHATVIHSGDYFLFESDSEEEEEALP  4569

Query  1434  edpRPSAQSAFQLAYQAWVTNaqavlrrrqqeqeqarqeqagqLPTGGGPSQEVepaegp  1493
             EDPRP+AQSAFQ+AYQAWVTN        +Q +E+ARQE+A QL +GG  + +VEP + P
Sbjct  4570  EDPRPAAQSAFQMAYQAWVTN---AQTVLRQRRERARQERAEQLASGGDLNPDVEPVDVP  4740

Query  1494  eeaaagRSHVVQRVLSTAQFLWMLGQALVDELTRWLQEFTRHHGTMSDVLRAERYLLTQE  1553
             E+  AGRSH++QRVLST QFLW+LGQA VD LTRWL+ FT+HH TMSDVL AERYLLTQE
Sbjct  4741  EDEMAGRSHMMQRVLSTMQFLWVLGQATVDGLTRWLRAFTKHHRTMSDVLCAERYLLTQE  4920

Query  1554  LLQGGEVHRGVLDQLYTSQAEATLPGPTEAPNAPSTVSSGLGAEEPLSSMTDDMGSPLST  1613
             LL+ GEV RGVLDQLY  + EATL GP E  + PST SSGLGAEEPLSSMTDD  SPLST
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            GY+TRSGSEE VTD G+ +AG SL+         +  MRTASELLLDRRL IPELEEAE
Sbjct  5101  GYNTRSGSEEIVTDAGDLQAGTSLHGSQELLANARTRMRTASELLLDRRLHIPELEEAER  5280

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            F   QGR LRLLRA YQCVAAHSELLCYFIIILNHMVTASA SLVLPVLVFLWAML+IPR
Sbjct  5281  FEAQQGRTLRLLRAGYQCVAAHSELLCYFIIILNHMVTASAASLVLPVLVFLWAMLTIPR  5460

Query  1725  PSKRFWMTAIVFTEIAVVVKYLFQFGFFPWNSHVVLRRYENKPYFPPRILGLEKTDGYIK  1784
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Sbjct  5461  PSKRFWMTAIVFTEVMVVTKYLFQFGFFPWNSYVVLRRYENKPYFPPRILGLEKTDSYIK  5640

Query  1785  YDLVQLMALFFHRSQLLCYGLWDHEEDSPSKEHDKSgeeeqgaee---------------  1829
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Sbjct  5641  YDLVQLMALFFHRSQLLCYGLWDHEEDRYPKDHCRSSVKDREAKEEPEAKLESQSETGTG  5820

Query  1830  --gpgVPAATTEDHIQVEARVGPTDGTPEPQVELRPrdtrrislrfrrrKKEGPARKGaa  1887
                 V A T  DHIQ +  +   D   +P  +L+P    R      RR+KE P  KG A
Sbjct  5821  HPKEPVLAGTPRDHIQGKGSIRSKDVIQDPPEDLKP-RHTRHISIRFRRRKETPGPKGTA  5997

Query  1888  aieaedreeeegeeekeaPTGREKrpsrsggrvraagrrLQGFCLSLAQGTYRPLRRFFH  1947
             +E E  E E  E  +       +  S+   R++AAGRRLQ FC+SLAQ  Y+PL+RFFH
Sbjct  5998  VMETEHEEGEGKETTERKRPRHTQEKSKFRERMKAAGRRLQSFCVSLAQSFYQPLQRFFH  6177

Query  1948  DILHTKYRAATDVYALMFLAdvvdfiiiifgfwafgKHSAATDITSSLSDDQVPEAFLVM  2007
            DILHTKYRAATDVYALMFLAD+VD IIIIFGFWAFGKHSAATDI SSLSDDQVP+AFL M
Sbjct  6178  DILHTKYRAATDVYALMFLADIVDIIIIIFGFWAFGKHSAATDIASSLSDDQVPQAFLFM  6357

Query  2008  LLIQFSTMVVDRALYLRKTVLGKLAFQVALVLAIHLWMFFILPAVTERMFNQNVVAQLWY  2067
            LL+QF TMV+DRALYLRKTVLGKLAFQV LV+AIH+WMFFILPAVTERMF+QN VAQLWY
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Query  2068  FVKCIYFALSAYQIRCGYPTRILGNFLTKKYNHLNLFLFQGFRLVPFLVELRAVMDWVWT  2127
            FVKCIYFALSAYQIRCGYPTRILGNFLTKKYNHLNLFLFQGFRLVPFLVELRAVMDWVWT
Sbjct  6538  FVKCIYFALSAYQIRCGYPTRILGNFLTKKYNHLNLFLFQGFRLVPFLVELRAVMDWVWT  6717

Query  2128  DTTLSLSSWMCVEDIYANIFIIKCSRETEkkypqpkgqkkkkivkygMGGliilfliaii  2187
            DTTLSLS+WMCVEDIYANIFIIKCSRETEKKYPQPKGQKKKKIVKYGMGGLIILFLIAII
Sbjct  6718  DTTLSLSNWMCVEDIYANIFIIKCSRETEKKYPQPKGQKKKKIVKYGMGGLIILFLIAII  6897

Query  2188  wfpllfMSLVRSVVGVVNQPIDVTVTLKLGGYEPLFTMSAQQPSIIPFTAQAYEELSRQF  2247
            WFPLLFMSL+RSVVGVVNQPIDVTVTLKLGGYEPLFTMSAQQPSI+PFT QAYEELS+QF
Sbjct  6898  WFPLLFMSLIRSVVGVVNQPIDVTVTLKLGGYEPLFTMSAQQPSIVPFTPQAYEELSQQF  7077

Query  2248  DPQPLAMQFISQYSPEDIVTAQIEGSSGALWRISPPSRAQMKRELYNGTADITLRFTWNF  2307
            DP PLAMQFISQYSPEDIVTAQIEGSSGALWRISPPSRAQMK+ELYNGTADITLRFTWNF
Sbjct  7078  DPYPLAMQFISQYSPEDIVTAQIEGSSGALWRISPPSRAQMKQELYNGTADITLRFTWNF  7257

Query  2308  QRDLAKGGTVEYANEKHMLALAPNSTARRQLASLLEGTSDQSVVIPNLFPKYIRAPNGPE  2367
            QRDLAKGGTVEY NEKH L LAPNSTARRQLA LLEG  DQSVVIP+LFPKYIRAPNGPE
Sbjct  7258  QRDLAKGGTVEYTNEKHTLELAPNSTARRQLAQLLEGRPDQSVVIPHLFPKYIRAPNGPE  7437

Query  2368  ANPVKQLQPNEEADYLGVRIQLRREQ-GAGATG---------FLEWWVIELQECRTDCNL  2417
            ANPVKQLQP+EE DYLGVRIQLRREQ G GA+G         FLEWWVIELQ+C+ DCNL
Sbjct  7438  ANPVKQLQPDEEEDYLGVRIQLRREQVGTGASGEQAGTKASDFLEWWVIELQDCKADCNL  7617

Query  2418  LPMVIFSDKVSPPSLGFLAGYGIMGLYVSIVLVIGKFVRGFFSEISHSIMFEELPCVDRI  2477
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             LPMVIFSDKVSPPSLGFLAGYGI+GLYVSIVLV+GKFVRGFFSEISHSIMFEELPCVDRI
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             LKLCQDIFLVRETRELELEEELYAKLIFLYRSPETMIKWTRE+E
Sbjct  7798  LKLCQDIFLVRETRELELEEELYAKLIFLYRSPETMIKWTRERE  7929
```

3. Chosen Protein Sequence

Chosen Protein:

```
>M. Musculus protein (sequence taken from BLAST result)
MEPHVLGAGLYWLLLPCTLLAASLLRFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRLLRALLCLSLLFLVAHLAFQI
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TWWSCHFPLSPLGFNTLCVMVSCFGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDLPNYSSPNALVLNTKHAW
PIYVSPGILLLLYYTATSLLKLHKSCPSELRKETPREDEEHELELDHLEPEPQARDATQGEMPMTTEPDLDNCTVHVLT
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CILLYGLTLCCLRYVWAMELPELPTTLGPVSLHQLGLEHTRYPCLDLGAMLLYLLTFWLLLRQFVKEKLLKKQKVPAAL
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LIFLYRSPETMIKWTRERE
```

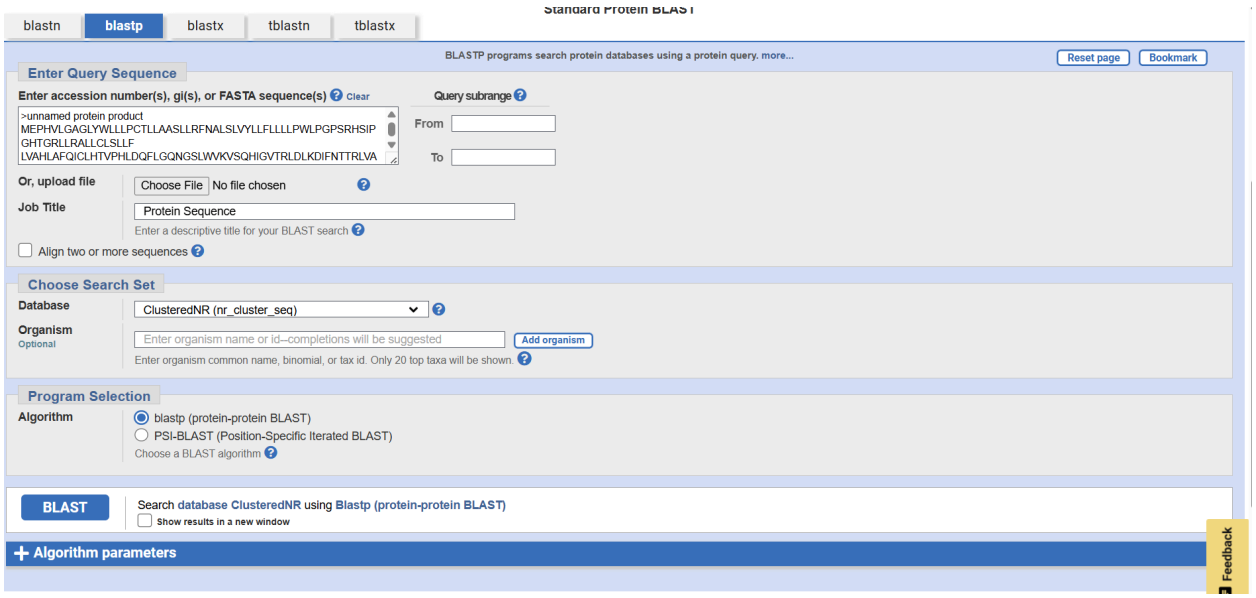
Name: piezo-type mechanosensitive ion channel component 1

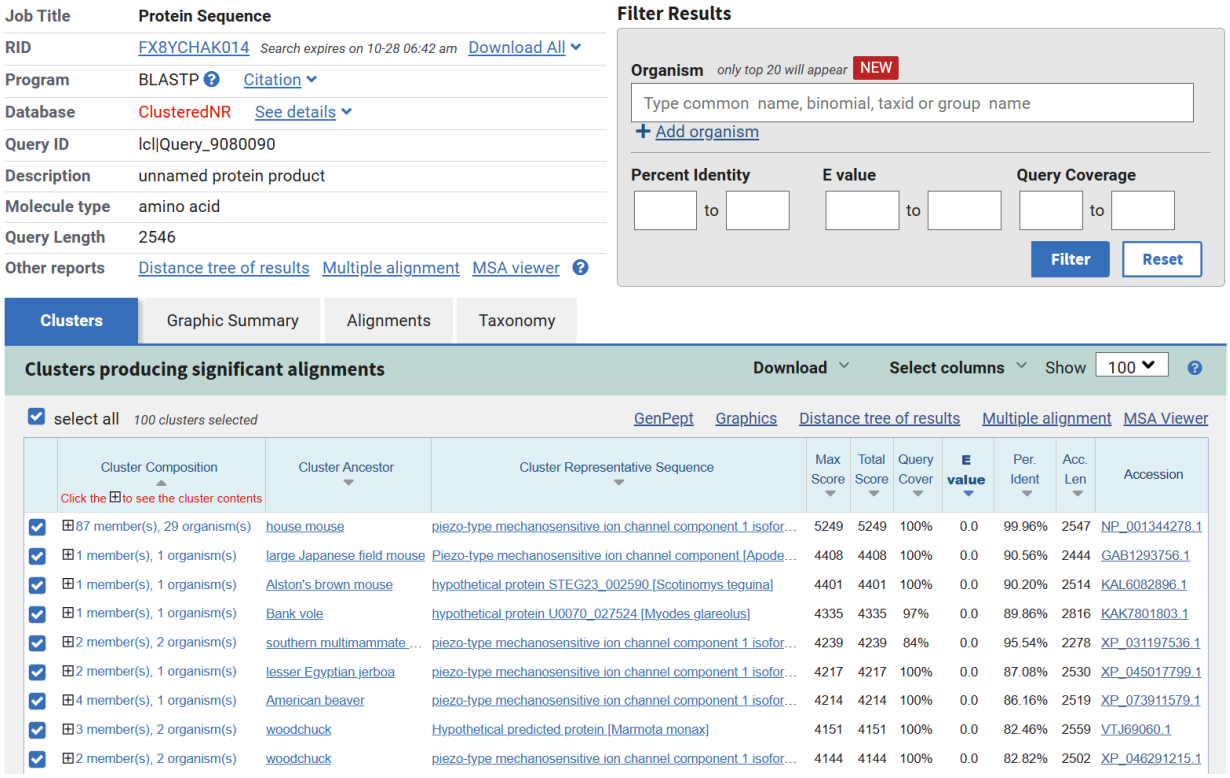
Species: Mus musculus

Eukaryota; Metazoa; Chordata; Craniata; Vertebrata; Euteleostomi; Mammalia; Eutheria; Euarchontoglires; Glires; Rodentia; Myomorpha; Muroidea; Muridae; Murinae; Mus; Mus

4. Novel Protein Identification

A BLASTP search against NR database (see setup in first screen-shot below) yielded a top hit result to a protein from House Mouse. See additional screenshots below for top hits and selected alignment details:





The top hit is from house mouse and has an E-value of 0.0, a 100% coverage, and 99.96% identity, which fulfills the criteria of a novel gene since the percent identity is less than 100%.

The alignment detail:

Download GenPept Graphics ▼ Next ▲ Previous ◀ Descriptions

piezo-type mechanosensitive ion channel component 1 isoform 1 [Mus musculus]

Sequence ID: NP_001344278.1 Length: 2547 Number of Matches: 1

Range 1: 1 to 2547 GenPept Graphics ▼ Next Match ▲ Previous Match

Score	Expect	Method	Identities	Positives	Gaps
5249 bits(13616)	0.0	Compositional matrix adjust.	2546/2547(99%)	2546/2547(99%)	1/2547(0%)

```
Query  1     MEPHVLGAGLYWLLLPCTLLAASLLRFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRLL  60
             MEPHVLGAGLYWLLLPCTLLAASLLRFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRLL
Sbjct  1     MEPHVLGAGLYWLLLPCTLLAASLLRFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRLL  60

Query  61    RALLCLSLLFLVAHLAFQICLHTVPHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTR  120
             RALLCLSLLFLVAHLAFQICLHTVPHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTR
Sbjct  61    RALLCLSLLFLVAHLAFQICLHTVPHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTR  120

Query  121   LVAPDLGVLLASSLCLGLCGRLTRKARQSRRTQEL-DDDDDDDDDDEDIDAAPAVGLKGA  179
             LVAPDLGVLLASSLCLGLCGRLTRKARQSRRTQEL DDDDDDDDDDEDIDAAPAVGLKGA
Sbjct  121   LVAPDLGVLLASSLCLGLCGRLTRKARQSRRTQELQDDDDDDDDDDEDIDAAPAVGLKGA  180

Query  180   PALATKRRLWLASRFRVTAHWLLMTSGRTLVIVLLALAGIAHPSAFSSVYLVVFLAICTW  239
             PALATKRRLWLASRFRVTAHWLLMTSGRTLVIVLLALAGIAHPSAFSSVYLVVFLAICTW
Sbjct  181   PALATKRRLWLASRFRVTAHWLLMTSGRTLVIVLLALAGIAHPSAFSSVYLVVFLAICTW  240

Query  240   WSCHFPLSPLGFNTLCVMVSCFGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDL  299
             WSCHFPLSPLGFNTLCVMVSCFGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDL
Sbjct  241   WSCHFPLSPLGFNTLCVMVSCFGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDL  300

Query  300   PNYSSPNALVLNTKHAWPIYVSPGILLLLYYTATSLLKLHKSCPSELRKETPREDEEHEL  359
             PNYSSPNALVLNTKHAWPIYVSPGILLLLYYTATSLLKLHKSCPSELRKETPREDEEHEL
Sbjct  301   PNYSSPNALVLNTKHAWPIYVSPGILLLLYYTATSLLKLHKSCPSELRKETPREDEEHEL  360

Query  360   ELDHLEPEPQARDATQGEMPMTTEPDLDNCTVHVLTSQSPVRQRPVRPRLAELKEMSPLH  419
             ELDHLEPEPQARDATQGEMPMTTEPDLDNCTVHVLTSQSPVRQRPVRPRLAELKEMSPLH
Sbjct  361   ELDHLEPEPQARDATQGEMPMTTEPDLDNCTVHVLTSQSPVRQRPVRPRLAELKEMSPLH  420

Query  420   GLGHLIMDQSYVCALIAMMVWSIMYHSWLTFVLLLWACLIWTVRSRHQLAMLCSPCILLY  479
             GLGHLIMDQSYVCALIAMMVWSIMYHSWLTFVLLLWACLIWTVRSRHQLAMLCSPCILLY
Sbjct  421   GLGHLIMDQSYVCALIAMMVWSIMYHSWLTFVLLLWACLIWTVRSRHQLAMLCSPCILLY  480

Query  480   GLTLCCLRYVWAMELPELPTTLGPVSLHQLGLEHTRYPCLDLGAMLLYLLTFWLLLRQFV  539
             GLTLCCLRYVWAMELPELPTTLGPVSLHQLGLEHTRYPCLDLGAMLLYLLTFWLLLRQFV
Sbjct  481   GLTLCCLRYVWAMELPELPTTLGPVSLHQLGLEHTRYPCLDLGAMLLYLLTFWLLLRQFV  540

Query  540   KEKLLKKQKVPAALLEVTVADTEPTQTQTLLRSLGELVTGIYVKYWIYVCAGMFIVVSFA  599
             KEKLLKKQKVPAALLEVTVADTEPTQTQTLLRSLGELVTGIYVKYWIYVCAGMFIVVSFA
Sbjct  541   KEKLLKKQKVPAALLEVTVADTEPTQTQTLLRSLGELVTGIYVKYWIYVCAGMFIVVSFA  600
```

Related Information

Gene - associated gene details

AlphaFold Structure - 3D structure displays

Genome Data Viewer - aligned genomic context

5. Multiple Sequence Alignment

Relabled Sequence Detail:

```
>M_Musculus protein (sequence taken from BLAST result)
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CILLYGLTLCCLRYVWAMELPELPTTLGPVSLHQLGLEHTRYPCLDLGAMLLYLLTFWLLLRQFVKEKLLKKQKVPAAL
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ERQRKLNYGTFGTNRTQGVNNAVKTPSRSQAVVDWLSSIFTTYWILIVALVLLLQALGER
ASFINIGFFIFWSLGVLQFKLSFRFFRGTVYASWTFLIIYTSAVIIAIYTYQFRSFPEFL
EKAGLSKDWAHNIGLYHYADDKSNTKSLFTRLLLPISLFVVTMLQLKFFHEPWSQAVQPI
RATSDSEPGTSNQAVSLSARLKAWAHYGSELLWRIMEVHIAKIVFITVMAIACKNITAMY
IPVVVLISLGLCLPRGATGLLGLILTAYLAVLALAKMVFQMSFIKADIFGDGNGTCEGDE
GENATSYNMSTTEWAGFKKNQNMFNELGGLLVSICALTLQSIVIYRQRNFRIRTSQPPNL
RHRIFPELDPVTYDHSLVNCFKFFVDFGFYKFGLECCLVMIAINAWIRMDMLGAVVIGWL
AVFALSKRTMARWLWPIFLIYLVLLLPIQFSIYVGLPPSLCIDYPWKNLFITNPTKRRNL
LIWLGLADYDIQWTSANLLADFFLLMFVSCQAAVFRVEGPTHPAGDNDSIYKDGVYQFKN
NPRYDFVTTQRSFVDYFKKFVFLYAQWMTLVTVMVAGLGGTSLFALGYLCLAFWILWSGN
ALYVMKNHNRTIKKFMIVLGYTVFSMFCKVSLQVVGCVFLGSLHGIPGKWGCILRQIFSI
ACVNNLAMEDIANLFPESTEFDKDCPVSAAETRIGFDTLALAFLVLQLRIFHSWYFQHAM
TDFRAEIVLANRGAVLANQLIEKEMKEQTQQQNEKFNDIKRRTDDIRAQYQKQQALERTF
EPKTYGQAKRAGDYYMFEKDPENLVPPIESFVPEVDPEGSEFRRLDPTQLVYTAVNKDMN
LSASMRQVEKAELIKDDEERRMIAAVTPDTPPTERVVDEPIAEESDSFFIATLKFVFKFA
YNVFDWTAVFLNRRSREHRYVAYVLGREKQRLKNEYGEILCDSTKPMADLQALVPLQQLQ
VVRTETDIEKLESEALGDWQKRSVFARLFNALGDCVNAHTDVICYVLAIMAHAYCAGLIT
LPLPLLVFFWGSLSNPRPSKMFWIVLIAITESIIIIKFIFQFGFFEFNTEQSQSLNNKNQ
YQWNKLFGVQRQSYYAFLDVTLLIFLFYHRYNLRRLGLWKDANINDTFSGNNTASIDPLE
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CIFIITFGYSSFGEGGSGNVVSDIQSNRIPLAFVIMLIVLTCMIVIDRGLYLRKAVYCKL
AYQLISIIFLHIWIFLVLPSITKREASSNRMALLLYFVKCMYFLVSAWQIRNGYPALCVG
NLLTHAYGLTNMVFFKIFMAVPFLFELRTAIDWTWTDTSMPLFDFFNMENFYATIYNLKC
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SISVDGYPPLYSIEAQGTELKHLTAAELSGLRSGFSVAFVPGPMQDDSIRKSRSAVSFLD
EYTQDDSLKVLFRPESEQSWDISDDSKTAMIAALNDTTTTMQMQIHISFTRPRQDSSKEP
VTHTSEFSIAMDNTTKIREQMLKALRDDTNSTVIVSNAIPVYLIVPNEGEVIQASALNWV
MNQFVKPELKQNSLATLKLTLNAQNLWTVEMQRPELNATNLLLPAEKVKYGTGGKEYLQL
VSFVDRTFPSFFAKYVQGGVIAMYIALVIVVGKVLRGLFTNQPLDVIIAEIPNPDHLLKI
CLDIYLVREAKDFVLEQDLFAKLIFLFRSPETLIKWTRYKFKDD

>Pan_troglodytes tr|H2QBQ5|H2QBQ5_PANTR Piezo type mechanosensitive ion channel component 1 OS=Pan troglodytes OX=9598 GN=PIEZO1 PE=3 SV=2
MEPHVLGAVLYWLLLPCALLAACLLRFSGLSLVYLLFLLLLPWFPGPTRCGLQGHTGRLL
RALLGLSLLFLVAHLALQICLHTVPHLDQLLGPSCSRWETLSRHIGVTRLDLKDIPNAIR
LVAPDLGILVVSSVCLGICGRLARNTRQSPHPRELDDDERDVDASPTAGLQEAATLAPTR
RSRLAARFRVTAHWLLVAAGRVLAVTLLALAGIAHPSALSSVYLLLFLATCTWWACHFPI
STRGFSRLCVAVGCFGAGHLICLYCYQMPLAQALLPPAGIWARVLGLKDFVGPTNCSSPH
ALVLNTSLDWPVYASPGVLLLLCYATASLRKLRAYRSSGQRKEAAKGYEARELELAELDQ
WPQERESDQHVVPTAPDTEADNCIVHELTGQSSLRQRPVRPKRAEPREASPLHSLGHLIM
DQSYVCALIAMMVWSITYHSWLTFVLLLWACLIWTVRSRHQLAMLCSPCILLYGMTLCCL
RYVWAMDLRPELPTTLGPVSLRQLGLEHTRYPCLDLGAMLLYTLTFWLLLRQFVKEKLLK
WAESPAALTEVTVADTEPTRTQTLLQSLGELVKGVYAKYWIYVCAGMFIVVSFAGRLVVY
KIVYMFLFLLCLTLFQVYYSLWRKLLKAFWWLVVAYTMLVLIAVYTFQFQDFPAYWRNLT
GFTDEQLGDLGLEQFSVSELFSSILVPGFFLLACILQLHYFHRPFMQLTDMEHVSLPGTR
LPRWAHRQDAVSGTPLLQEEQEEQQQEEEEDSRNEGLGVATPHQATQVPEGAAKWGLVAE
RLLELAAGFSDVLSRVQVFLRRLLELHVFKLVALYTVWVALKEVSVMNLLLVVLWAFALP
YPRFRPMASCLSTVWTCVIIVCKMLYQLKVVNPQEYSSNCTEPSPNSTNLLPTEISQSLL
YRGPVDPANWFGVRKGFPNLGYIQNHLQVLLLLVFEAIVYRRQEHYRRQHQLAPLPAQAV
FASGTRQQLDQDLLGCLKYFINFFFYKFGLEICFLMAVNVIGQRMNFLVILHGCWLVAIL

TRRHRQAIARLWPNYCLFLALFLLYQYLLCLGMPPALCIDYPWRWSRAVPMNSALIKWLY
LPDFFRAPNSTNLISDFLLLLCASQQWQVFSAERTEEWQRMAGVNSDRLEPLRGEPNPVP
NFIHCRSYLDMLKVAVFRYLFWLVLVVVFVTGATRISIFGLGYLLACFYLLLFGTALLQR
DTRARLMLWDCLILYNVTVVISKNMLSLLACVFVEQMQTGFCWVIQLFSLVCTVKGYYDP
KEMMDRDQDCLLPVEEAGIIWDSVCFFFLLLQRRVFLSHYYLHVRADLQATALLASRGFA
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PGPDSPGGSSPPRRQWWRPWLDHATVIHSGDYFLFESDSEEEEEAVPEDLRPSAQSAFQL
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VLSTAQFLWMLGQALVDGLTRWLQEFTRHHGTMSDVLRAERYLLTQELLQGGEVHRGVLD
QLYTSQAEATLPGPTEAPNAPSTVSSGLGAEEPLSSMTDDTGSPLSTGYHTRSGSEEAVT
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SELFCYFIIILNHMVTASAGSLVLPVLVFLWAMLSIPRPSKRFWMTAIVFTEIAVVVKYL
FQFGFFPWNSHVVLRRYENKPYFPPRILGLEKTDGYIKYDLVQLMALFFHRSQLLCYGLW
DHEEESPSKEHDKSGEEEQGAEEGPGVPAATTKDHVQVEARVGPTDGTPEPQVELRPRDT
RRISLRFRRRKKEEGPARKGAAAIEAEDREEEEGEEEKEAPTGREKRPSRSGGRVRAAGR
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FQGFRLVPFLVELRAVMDWVWTDTTLSLSSWMCVEDIYANIFIIKCSRETEKRYPQPKGQ
KKKKIVKYGMGGLIILFLIAIIWFPLLFMSLVRSVVGVVNQPIDVTVTLKLGGYEPLFTM
SAQQPSIIPFTAQAYEELSRQFDPQPLAMQFISQYSPEDIVTAQIEGSSGALWRISPPSR
AQMKRELYNGTADITLRFTWNFQRDLAKGGTVEYANEKHMLALAPNSTARQQLASLLEGT
SDQSVVIPNLFPKYIRAPNGPEANPVKQLQPNEEADYLGVRIQLRREQGAGATGFLEWWV
IELQECRTDCNLLPMVIFSDKVSPPSLGFLAGYGIMGLYVSIVLVIGKFVRGFFSEISHS
IMFEELPCVDRILKLCQDIFLVRETRELELEEELYAKLIFLYRSPETMIKWTREKE

>Rhinopithecus_roxellana tr|A0A2K6PRC5|A0A2K6PRC5_RHIRO Piezo type mechanosensitive ion channel component 1 (Er blood group) OS=Rhinopithecus roxellana OX=61622 GN=PIEZO1 PE=3 SV=1
MRTEAYGGFRGSDPGQHSQGTWPRGCDELGRGCAVLWGGGWPSSVGDAADVHLPAACLLR
FSGLSLVYLLFLLLLPWFPGPTRHSLQGHTGRLLRALLGLSLLFLVAHLALQICLHTVPH
LDQLLGPSCSRWESLSRHIGVTRLDLKDVPNTIRLVAPDLGILVVSSVCLGLCGRLARNT
QQSPHTQELVREAASPVCQGGHSRALVSAILRVNLLKTNVDFLVVSCQVQPCVPLPCPAG
IAHPSALSSVYLLVFLAICTWWACHFPISTLGFNRLCVAVGCFGAGHLICLYCYQTPLAQ
ALLPPAGIWARVLGLKDFVGPANCSSPHVLVLNTSLDWPIYASPGILLLLCYATASLRKL
RVHRPSGQRKEAAKGDETRELELAELDQWPQDGEAAQHVVPTAPDTEADNCIVHELTGQS
PIRQRPVQPKRAEPREASPLHSLGHLIMDQSYVCALIAMMVWSITYHSWLTFVLLLWACL
IWTVRSRHQLAMLCSPCILLYGMTLCCLRYVWAMDLRPELPTTLGPVSLRQLGLEHTRYP
CLDLGAMLLYTLTFWLLLRQFVKEKLLKRAEAPAALTEVTVADTEPTRTQTLLQSLGELV
KGVYAKYWIYVCAGMFIVVSFAGRLVVYKIVYMFLFLLCLTLFQVYYSLWRKLLKAFWWL
VVAYTMLVLIAVYTFQFQDFPAYWRNLTGFTDEQLGDLGLEQFSVSELFSSILVPGFFLL
ACILQLHYFHRPFMQLTDMEHVSPPGRRLPRWAHRQDAVSGTPLLQQQQEEEEEEEGPRA
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YTVWVALKEVSVMNLLLVVLWAFALPYPRFRPMASCLSTVWTCVIIVCKMLYQLKVVHPQ
EYSSNCTEPSPNSTNLLTTEISQSLLYRGPVDPANWFGVRKGFPNLGYIQNHLQVLLLLV
FEAIVYRRQEHHRRQHQLAPLPAQAVFASGTRQQLDQDLLGCLKYFINFFFYKFGLEICF
LMAVNVIGQRMNFMVTLHGCWLVAILTRRHRQAIARLWPNYCLFLALFLLYQYLLCLGMP
PALCIDYPWRWSRAVPMNSALIKWLYLPDFFRSPNSTNLISDFLLLLCASQQWQVFSAER
TEEWQCMAGVNTDRLEPLRGEPNPVPNFIHCRSYLDMLKVAVFRYLFWLVLVVVFVTGAT
RISIFGLGYLLACFYLLLFGTALLQRDTRARLVLWDCLILYNVTVIISKNMLSLLACVFV
EQMQTGFCWVIQLFSLVCTVKGYYDPKEMMGRDQDCLLPVEEAGIIWDSICFFFLLLQRR
VFLSYYYLHVRADLQATALLASRGFALYNAANLKSIDFHRRTEEKSLAQLKRQMERIRAK
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FESDSEEEEEAAPEDPRPSAQSAFQMAYQAWVTNAQTVLRRQQQEQARQEHAGQLPTGGP
SQEVEPAEGPEEAVAGRSHMVQRVLSTAQFLWVLGQALVDGLTRWLQEFTRHHSTMSDVL
RAERYVLTQELLQGGEVHRGILDQLYTSEAEATLPGPTEAPDAPSTVSSGLGAEEPLSSM
TDDTGSPLSTGYHTRSGSEEAVTDPGEREAGASLHQGRMRTASELLLDRRLRIPELEEAQ

LFAEGQGRALRLLQAVYQCVAAHSELLCYFIIILNHMVTASAGSLVLPVLVFLWAMLSIP
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YLRKTVLGKLAFQVALVLAIHLWMFFILPAVTERMFSQNAVAQLWYFVKCIYFALSAYQI
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GVVNQPIDVTVTLKLGGYEPLFTMSAQQPSIIPFTAQAYEELSRQFDPHPLAMQFISQYS
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GLYVSIVLVIGKFVRGFFSEISHSIMFEELPCVDRILKLCQDIFLVRETRELELEEELYA
KLIFLYRSPETMIKWTREKE

>Tupaia_chinensis tr|L9KK63|L9KK63_TUPCH Protein PIEZO1 OS=Tupaia chinensis
OX=246437 GN=TREES_T100001696 PE=4 SV=1
MVCVWSCSDTFSHFCAGPKDMGLVPLTPGHTGRLLRALLCLSLLFLVAHLALQICLYTVP
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ARQRRHAQEPDSEEREVDGSALAGPQGAPVPAPKRKSELAARFRVTAHWLLVAAGRTLAI
VLLALAGIAHPSALSSVYLLVFLAVCTWWACHFPISPRGFSTLCVVVGCYSAGHLVCLYC
YQTPFAQAVLPPAGVWARVLGLKDFVGPVNCSSPNTLVLNTGHDWPVYVSPGILLLLYYT
VTSLLKLRGPRPADQRNEAARGDEERELELAELAQWPQGHTVAVQDGEATQPSTEADNCI
VHDLTGQSQRPAGQRPARPGLAEPREAPALHGLGQLIVEQSYVCALIAMMVSSRRRPWAC
GVWGGGPWQAGPRSTAGRVCVCTRGLSRPASQVWSITFHSWLTLVLLLWACLLWMARSRR
QLALLCSPCLLLYGLALCGLRYVWALDLRPELPTALGPVGLRQLGLEHTRYPCLDLGAML
LCTLTFWLLLRQFVREKLLRNARGTAAPSELAMVDGEPSQTRTLLRSLGVLVQGMYAKYW
IYVCAGMFIVVSFAGRLVVYKIVYMLLFLLCLTLFXXXXXXXXXXXXXXXXXXXXXXXXX
XXXXXXXXXXXXXXGLVVAYTMLVLIAVYTFQFQDFPAYWRNLTGLTDEQLGDLGLEQFS
VSELFSSILIPGFFLLACILQLHYFHTPFMQLTDPHRPRPPGARLLRWAPRQDTGSAALL
LQQEDQAPLDAGLRRAGPRQAMQAPEGASSKWGLVAERLLDLAASFSDALGRVQVLVRRL
LELHVFKLVALYTVWVALREVSVMNLLLLVLWAFALPYPRFRPMASCLCTVWTCVIIVCK
MLYQLRAVSPHAYSSNCSQPFPNSTNLQPAELSQSLLYRGPVDPANWFGVRKGFPNLGYI
QSHLQILLLLVFEAMVYRRQEHYRRQHQLGPPPARSVCVDGTRQRLDQGVLSCLKYFINF
FFYKFGLEICFLMAVNVIGQRMDFMAILHGCWLGAILTRRRRVAIASLWPNYCLFLTLFL
LYQYLLCLGIPPALCLDYPWRWSRAVPMNSALIKWLYLPDFFRAPNSTNLIIRAPSSEWA
RPTQQPRHVWAGLWVCADVRLGQHVCRGVSSAYWDMPLLSCVFVEQMQSSFCWVIQLFSL
VCTVKGYYDPQATTGRDLDCLLPVEEAGVLWDSICFSFLLLQRRVFLSHYFLHVRADLQA
TALQASRGFALYNAASLKRIDSRRKAEEKSLAQLKRQMERIRTKQQKHRQGGPGRGRPQD
PLGPEEPGQEPGPGSPGGSSPPRRQWWRPWLDHATVIHSGDYFLFESDSEEDEEEEEEAL
PEDPRPSGQSAFQMAYQAWVTSAQTVLRRQERGRQAEQLPVGGGPSQEVEPAEGPGDEPA
GRSHVVQRVLSMAQFLWVLGQALVDGLTRWLQAFTRHHGTMSDVLRAERYLLTQQLLQGG
EVDRGVLDQLYASEAGDTQAGTSEARDAPSTVSSGLGAEEPLSSLTEDTSSPLSTGYNTR
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GRALRLLRGVVWGGAAHPSGRGRGGWGGRGRRPVLSLLCYGLWDHEEDPLSKEPGGRKEL
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IWFPLLFMSLVRSVVGVVNQPIDVTVTLKLGGYEPLFTMSAQLQGPWVEGHFQLGALPTP
RPASPQPLFTMSAQLQGPWVEGHFQLGALPMPRPASPQPLFTMSAQQPSIVPFTPQAYEQ
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LTWNFQRDLAKGGTVEYSNDKHTLDLAPNSTARRQLASLLEGASDQSVVMGLYVSIVLVI
GKFVRGFFSEISHSIMFEELPCVDRILKLCQDIFLVRETRELELEEELYAKLIFLYRSPE

TMIKWTREKE

Alignment: Obtained using MUSCLE (version 3.8) at EBI:

```
CLUSTAL multiple sequence alignment by MUSCLE (3.8)

Panicum_hallii
MARRTGAAGFL------------------------------LPLV--------LLAASLL
Herrania_umbratica
--MGSFLAGFV------------------------------LPLL--------LLTAALI
Mesorhabditis_belari
-MECERKESLAASPKENIVKKWLLEMVSPLLK--TITYRGLLPAA--------LIAAAFI
Tupaia_chinensis
-MVCV-------------------------------------WSCS-----------DTFS
M_Musculus
-MEPHVLGAGL----------YW------LL----------LPCT--------LLAASLL
Pan_troglodytes
-MEPHVLGAVL----------YW------LL----------LPCA--------LLAACLL
Rhinopithecus_roxellana
-MRTEAYGGFRGSDPGQHSQGTWPRGCDELGRGCAVLWGGGWPSSVGDAADVHLPAACLL
                                                                .
:

Panicum_hallii
DWNLISLTNMIIFFAIR-FVAPTR----GFRTWR-LYLLLWCTIIYSVLAILAQVTFHII
Herrania_umbratica
NWSLVSLVDLIAFLLIQ-YTAPKI----GFRFRR-KYLLLWPVIIFSLLVFLSQAVYLVI
Mesorhabditis_belari
RPSFISVIYLVLALIGPVLPSTSRGAPLNVTVKFYVYIVFLVTILTTIAQCAFQI-YEST
Tupaia_chinensis
HFCA-GPKDMGLVPLTP-----------GHTGRL-LRALLCLSLLFLVAHLALQICLYTV
M_Musculus
RFNALSLVYLLFLLLLPWLPGPSRHSIPGHTGRL-LRALLCLSLLFLVAHLAFQICLHTV
Pan_troglodytes
RFSGLSLVYLLFLLLLPWFPGPTRCGLQGHTGRL-LRALLGLSLLFLVAHLALQICLHTV
Rhinopithecus_roxellana
RFSGLSLVYLLFLLLLPWFPGPTRHSLQGHTGRL-LRALLGLSLLFLVAHLALQICLHTV
                                  .   :                 .   .     ::   ::
:     *

Panicum_hallii
WCIEGMGWSVAYSW--WVKLVGLARGQPWESPSVIYFLALQLSATV-----LALVEVLGS
Herrania_umbratica
WAADGYKQSVGDAW--WMKLIGFMIIQSWKSPTVIYFLVVQLLVVF-----VALLDIHGT
Mesorhabditis_belari
RTIKTTSDCSDDLYR-WLRQAGFLRYVADTGFDSWRSFLPEVFSLICSFCTLIIVQCLNH
Tupaia_chinensis
PRLKELLEPNCSPWETLSRHVGVTRLDLKDIPNAIRLVAPDLGVLVASSICLGLCGRLAR
```

```
M_Musculus
PHLDQFLGQNGSLWVKVSQHIGVTRLDLKDIFNTTRLVAPDLGVLLASSLCLGLCGRLTR
Pan_troglodytes
PHLDQLLGPSCSRWETLSRHIGVTRLDLKDIPNAIRLVAPDLGILVVSSVCLGICGRLAR
Rhinopithecus_roxellana
PHLDQLLGPSCSRWESLSRHIGVTRLDLKDVPNTIRLVAPDLGILVVSSVCLGLCGRLAR
                                .        :    .  *.              .  ::
.     : :

Panicum_hallii
RL----H------------------QDSCWLNFSFGFEQIGYHLRVTCCF----------
Herrania_umbratica
KFGLVPW------------------RYSCWGHFLTAVEHLGSHLRVASCL----------
Mesorhabditis_belari
QV-FIEHPAGVHTVRTNSNNTQNATQRSNVEVLSRALLIALKRLSNAALF----------
Tupaia_chinensis
KARQRRHAQEP------DSEEREVDGSALAGPQGAPVPAPKRKSELAARFRVTAHWLLVA
M_Musculus
KARQSRRTQELDDDDDDDDDDEDIDAAPAVGLKGAPALATKRRLWLASRFRVTAHWLLMT
Pan_troglodytes
NTRQSPHPREL------DDDERDVDASPTAGLQEAATLAPTRRSRLAARFRVTAHWLLVA
Rhinopithecus_roxellana
NTQQSPHTQEL---------VREAASPVCQGGHSRALVSAILRVNL---LKTNVDFLVVS
                            .     .                                  .
:

Panicum_hallii
----LLPAAQLVVSISHPSWIS---LPFFVFSCIGLVDW---SLTSNFLGLFRWWRLLEM
Herrania_umbratica
----LLPPIQLVVGISHPSWVS---LPFFIGSCVGLVDW---SLTSNFLGLFRLWKALQL
Mesorhabditis_belari
-------IFVALAGCIQPSILNSFYFLIFLFMAT----WWSTYKPLRHSIYNKMKITLLF
Tupaia_chinensis
AGRTLAIVLLALAGIAHPSALSSVYLLVFLAVCT----WWACHFPISPRGFSTLCVVVGC
M_Musculus
SGRTLVIVLLALAGIAHPSAFSSVYLVVFLAICT----WWSCHFPLSPLGFNTLCVMVSC
Pan_troglodytes
AGRVLAVTLLALAGIAHPSALSSVYLLLFLATCT----WWACHFPISTRGFSRLCVAVGC
Rhinopithecus_roxellana
CQVQPCVPLPCPAGIAHPSALSSVYLLVFLAICT----WWACHFPISTLGFNRLCVAVGC
                                          ..  :** ..   : .*:  .     *     .
:

Panicum_hallii
YSVFSILLLYIYQLPVQFPYV--------VLAFADFIG--------LFKVSSKSEWPELS
Herrania_umbratica
YAGFNIVLLYVYQLPIEFSNM--------LQRIADFVG--------LFKISTASEWPEIC
Mesorhabditis_belari
YSGLHLILIYLYQIPEISKLLPPDQTAARMIGFSQVLK-TDCSMFYHIEINNALAWNVAI
Tupaia_chinensis
YSAGHLVCLYCYQTPFAQAVLPPAGVWARVLGLKDFVGPVNCSSPNTLVLNTGHDWPVYV
M_Musculus
FGAGHLICLYCYQTPFIQDMLPPGNIWARLFGLKNFVDLPNYSSPNALVLNTKHAWPIYV
```

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Pan_troglodytes
FGAGHLICLYCYQMPLAQALLPPAGIWARVLGLKDFVGPTNCSSPHALVLNTSLDWPVYA
Rhinopithecus_roxellana
FGAGHLICLYCYQTPLAQALLPPAGIWARVLGLKDFVGPANCSSPHVLVLNTSLDWPIYA
                            :.   :: :* ** *     :        :  : :.:
: :..   *

Panicum_hallii
SGISLLVYYFMLSSV-----------KQDIQETDSLISLEDGSL----------TEHLLP
Herrania_umbratica
SAVSLILFYIMLSYV-----------KCDLEEMDFIMSMRESKL----------TEQLLP
Mesorhabditis_belari
NPLIITGFYFLI--VLQLN-WTSSGTRNYLDDEDASSSVH-EEIQGVDENTNSEVEH--P
Tupaia_chinensis
SPGILLLLYYTVTSLLKLRGPRPADQRNEAARGDEERELELAELAQWPQGHTVAVQDGEA
M_Musculus
SPGILLLLYYTATSLLKLHKSCPSELRKETPREDEEHELELDHLEPEPQARDA-TQGEMP
Pan_troglodytes
SPGVLLLLCYATASLRKLRAYRSSGQRKEAAKGYEARELELAELDQWPQERES-DQHVVP
Rhinopithecus_roxellana
SPGILLLLCYATASLRKLRVHRPSGQRKEAAKGDETRELELAELDQWPQDGEA-AQHVVP
                            .   :        :          .         .:    :
:   .

Panicum_hallii
SGN-----------------AFFSCQSRADRRHTNI---------LLRGSVFRNFSINFF
Herrania_umbratica
SKH-----------------SFFIRESRSGVRHTNV---------LLRRTVFRTFTINFF
Mesorhabditis_belari
LVGDRETQSTIVPSGSSNSSTNIQSQSSPPTRQQRLLDTEERSPANRESVPMRKITSQVI
Tupaia_chinensis
TQPSTEADNCIV--------HDLTGQSQRPAGQRPA---RPGLAEPREAPALHGLGQLIV
M_Musculus
MTTEPDLDNCTV--------HVLTSQS--PVRQRPV---RPRLAELKEMSPLHGLGHLIM
Pan_troglodytes
TAPDTEADNCIV--------HELTGQS--SLRQRPV---RPKRAEPREASPLHSLGHLIM
Rhinopithecus_roxellana
TAPDTEADNCIV--------HELTGQS--PIRQRPV---QPKRAEPREASPLHSLGHLIM
                                                     :  :*     :
:. :   ..

Panicum_hallii
TY--------------------------------GFPVLLLALA----------------
Herrania_umbratica
TY--------------------------------GFPVSLFALS----------------
Mesorhabditis_belari
DKQKISQIFNAPGNEQSGTTAQTIVALIAFCVEHSYVFGLMTMM----------------
Tupaia_chinensis
EQ--------------------------------SYVCALIAMMVSSRRRPWACGVWGGG
M_Musculus
DQ--------------------------------SYVCALIAMM----------------
Pan_troglodytes
DQ--------------------------------SYVCALIAMM----------------
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Rhinopithecus_roxellana
DQ--------------------------------SYVCALIAMM----------------
                                                                .:   *:::

Panicum_hallii
--------------------------FWSFNFTSICAFGLLAYVGYILYAFPSLFQMHRL
Herrania_umbratica
--------------------------FWSFHFASICAFGLLAYVGYIVYAFPSLFRLHRL
Mesorhabditis_belari
--------------------------IWALLYHSAFGLGLLAW-ACIIWMFRDTRSVSFK
Tupaia_chinensis
PWQAGPRSTAGRVCVCTRGLSRPASQVWSITFHSWLTLVLLLW-ACLLWMARSRRQLALL
M_Musculus
--------------------------VWSIMYHSWLTFVLLLW-ACLIWTVRSRHQLAML
Pan_troglodytes
--------------------------VWSITYHSWLTFVLLLW-ACLIWTVRSRHQLAML
Rhinopithecus_roxellana
--------------------------VWSITYHSWLTFVLLLW-ACLIWTVRSRHQLAML
                                                      .*:: : *   : ** : .
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Panicum_hallii
-NGSLLVFVLLWAASTYVFNVAF-TFFNKRFQKDMK--IWETVGLWH--YSIPGLFLLAQ
Herrania_umbratica
-NGLLLVFILLWAVSTYIFNVAF-AFLNRNFGKDME--IWEMVGFWH--YPIPGFFLLAQ
Mesorhabditis_belari
CSPFIVSYVEFLLVIQYVWSMDL-EFLPAKAQTETKANVLEIIGFIN--YSGSDSFLVTF
Tupaia_chinensis
CSPCLLLYGLALCGLRYVWALDLRPELPTALGP--V--GLRQLGLEHTRYPCLDLGAMLL
M_Musculus
CSPCILLYGLTLCCLRYVWAMEL-PELPTTLGP--V--SLHQLGLEHTRYPCLDLGAMLL
Pan_troglodytes
CSPCILLYGMTLCCLRYVWAMDLRPELPTTLGP--V--SLRQLGLEHTRYPCLDLGAMLL
Rhinopithecus_roxellana
CSPCILLYGMTLCCLRYVWAMDLRPELPTTLGP--V--SLRQLGLEHTRYPCLDLGAMLL
                              .  :: :        *:: : :   :              :*:
:  *.  .   :

Panicum_hallii
FCLGVFVALCNLVNNSVFLCVQTVDGASS--------------SDDHLIDEKEDTMVLIV
Herrania_umbratica
FCLGILVALGNLVNNSVFVYSSDEDALSS--------------NNNSAVEVDGETKVFIV
Mesorhabditis_belari
VKV--LLSLPIFLLHRLKRRDSHYSRLTEHERQRKLNYGTFGTNRTQGVNNAVKTPSRSQ
Tupaia_chinensis
CTLTFWLLLRQFVREKLLRNARGTAAPSE-------------LAMVDGEPSQTRTLLRSL
M_Musculus
YLLTFWLLLRQFVKEKLLKKQKVPAALLE-------------VTVADTEPTQTQTLLRSL
Pan_troglodytes
YTLTFWLLLRQFVKEKLLKWAESPAALTE-------------VTVADTEPTRTQTLLQSL
Rhinopithecus_roxellana
YTLTFWLLLRQFVKEKLLKRAEAPAALTE-------------VTVADTEPTRTQTLLQSL
                                :   : *  ::   :           .
*
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Panicum_hallii
ATLAW--GLRKLSRAITLTLLFLLVMKRGFIHAVYMCFFLVFLVNHSIDKRLRQILVLFS
Herrania_umbratica
ATIAW--GLCKCSRAIMLALIFVIAMKPGFIHAVYVIFFLIYLLSHNISRKIRQSLILLC
Mesorhabditis_belari
AVVDWLSSIFTTYWILIVALVLLLQALGERASFINIGFFIFW---------------SLG
Tupaia_chinensis
GVLVQ--GMYAKYWIYVCAGMFIVVSFAGRLVVYKIVYMLLF---------------LLC
M_Musculus
GELVT--GIYVKYWIYVCAGMFIVVSFAGRLVVYKIVYMFLF---------------LLC
Pan_troglodytes
GELVK--GVYAKYWIYVCAGMFIVVSFAGRLVVYKIVYMFLF---------------LLC
Rhinopithecus_roxellana
GELVK--GVYAKYWIYVCAGMFIVVSFAGRLVVYKIVYMFLF---------------LLC
                            . :    .:    .    : ::::           : ::::.:
:

Panicum_hallii
EVHFSILYILQLDLVSSALERSGSLTMEVLSQLGLSNNATTKDFIEIGSIVCFCAVHSHG
Herrania_umbratica
EAHFALLYLLQIELISNALEQKGSLSLEIILQLGLLKHNSSWEFLEIALLACFCAIHNHG
Mesorhabditis_belari
VLQFKLSFRFFRGTVYAS----------------------WT------------------
Tupaia_chinensis
LTLFXXXXXXXXXXXXXX----------------------XXXXXXXXXXXXXXXXXXXX
M_Musculus
LTLFQVYYTLWRKLLRVF----------------------WW------------------
Pan_troglodytes
LTLFQVYYSLWRKLLKAF----------------------WW------------------
Rhinopithecus_roxellana
LTLFQVYYSLWRKLLKAF----------------------WW------------------
                                *

Panicum_hallii
FKMLFSLSAVLRHTPCPPVGFTILKAGLNKSVLLSVYSSQSSRNDEACRN--SHERKIAS
Herrania_umbratica
FEMLFSFSAIVQHTPSPPVGFSILRAGLNKSVLLSVYASPN---PSYCHDNASYERRIAA
Mesorhabditis_belari
------FL----------IIYTSA-------VIIAIYTYQFRSFPEFLEK-AGLSKD---
Tupaia_chinensis
XXXXXGLV----------VAYTML-------VLIAVYTFQFQDFPAYWRNLTGLTDE---
M_Musculus
------LV----------VAYTML-------VLIAVYTFQFQDFPTYWRNLTGFTDE---
Pan_troglodytes
------LV----------VAYTML-------VLIAVYTFQFQDFPAYWRNLTGFTDE---
Rhinopithecus_roxellana
------LV----------VAYTML-------VLIAVYTFQFQDFPAYWRNLTGFTDE---
                                   :           : ::         *::::*:
.  .

Panicum_hallii
YLSKIGQKFLSVYRSYGTYVAFLTILLTLYLVTPNYISFGYLFFLLFWIIGRQLVEKTKR
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Herrania_umbratica
FLSEIGQTFLSIYRSCGTYIALLTILLTVYMVTPNYISFGYIFLLLVWITGRQLVERTKR
Mesorhabditis_belari
WAHNIGLYHYADDKS-NTKSLFTRLLLPISLFVVTMLQLKFFHEP---------------
Tupaia_chinensis
QLGDLGLEQFSVSE------LFSSILIPGFFLLACILQLHYFHTPFMQLTDPHRPRPPGA
M_Musculus
QLGDLGLEQFSVSE------LFSSILIPGFFLLACILQLHYFHRPFMQLTDLEHVPPPGT
Pan_troglodytes
QLGDLGLEQFSVSE------LFSSILVPGFFLLACILQLHYFHRPFMQLTDMEHVSLPGT
Rhinopithecus_roxellana
QLGDLGLEQFSVSE------LFSSILVPGFFLLACILQLHYFHRPFMQLTDMEHVSPPGR
                                .:*    :          :  :*:.  :.    :.: ::.

Panicum_hallii
RL--WFPLKVYAAMVFIFTYSLSV--SPIFAESVSEFVKLYPDLGFDPKASLLENVWQSL
Herrania_umbratica
RL--WFPLKTYAIMVFIFVYSLSS--FTSFKIWLSSFVDLYFYLGYDPEGSLLDNIWQSL
Mesorhabditis_belari
----WSQ-----AVQPIRATSDSEPGTSNQAVSLSARLKAWAHYGS-------ELLW---
Tupaia_chinensis
RLLRWAPRQDTGSAALLL--------QQEDQAPLDAGLRRAGPRQAMQAPEGASSKWGLV
M_Musculus
RHPRWAHRQDAVSEAPLLEH------QEEEEVFREDGQSMDGPHQATQVPEGTASKWGLV
Pan_troglodytes
RLPRWAHRQDAVSGTPLLQEEQEEQQQEEEEDSRNEGLGVATPHQATQVPEG-AAKWGLV
Rhinopithecus_roxellana
RLPRWAHRQDAVSGTPLLQQQQEE--EEEEEGPRAEGLGMATPHQATQVPEG-AAKWGLV
                                 *          :
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Panicum_hallii
AVLVVMQLYSYERRQNSDKNFGVSDALVSGLLGFLRRFLIWHSDKMLSLSVFYACLSSIS
Herrania_umbratica
AVLIVMQLYSYERRQSKYNRTDDPNPLDSGLLGFAKRFLIWHSQKVLFVSLFYASISPIS
Mesorhabditis_belari
------------------------------------RIMEVHIAKIVFITVMAIACKNIT
Tupaia_chinensis
AERLLDLAASF-------------SDALGRVQVLVRRLLELHVFKLVALYTVWVALREVS
M_Musculus
ADRLLDLAASF-------------SAVLTRIQVFVRRLLELHVFKLVALYTVWVALKEVS
Pan_troglodytes
AERLLELAAGF-------------SDVLSRVQVFLRRLLELHVFKLVALYTVWVALKEVS
Rhinopithecus_roxellana
AERLLELAAGF-------------SDVLSRVQVFLRRLLELHVFKLVAVYTVWVALKEVS
                                                                 *::  *
*:: :  .  .   ::

Panicum_hallii
LSGLIYLLGLIVFCTLPK-MSRIPSKVYLVYTGLLAASEYLFQMVCKPAQMCPDQHFYGL
Herrania_umbratica
ACGFLYLLGLVICSTLPK-ASRIPSKSFLLYTGFLMTTEYLYQMWGKQAGMFPGQKHSDL
Mesorhabditis_belari
AMYIPVVVLISLGLCLPRGATGLLGLILTAYLAVLALAKMVFQMSFIKADIFGDGN----
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Tupaia_chinensis
VMNLLLLVLWAFALPYPR-FRPMASCLCTVWTCVIIVCKMLYQLRAVSPHAYSSN-----
M_Musculus
VMNLLLVVLWAFALPYPR-FRPMASCLSTVWTCIIIVCKMLYQLKIVNPHEYSSN-----
Pan_troglodytes
VMNLLLVVLWAFALPYPR-FRPMASCLSTVWTCVIIVCKMLYQLKVVNPQEYSSN-----
Rhinopithecus_roxellana
VMNLLLVVLWAFALPYPR-FRPMASCLSTVWTCVIIVCKMLYQLKVVHPQEYSSN-----
                              :  ::   .    *.    : .     :  .:   : ::*:
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Panicum_hallii
SVFLGLKYYDSGFWGVEYGLRGKVLVIVACTIQYNVFHWLDMMPTSLVHNGKWEEPCQLF
Herrania_umbratica
SLFLGFRVYELGFWGIESGLRGKVLVIAACIFQYNVFHWLDNMPSGISSKGKWEEPCPLF
Mesorhabditis_belari
---------------------------GTCEGDEGENATSYNMSTTE-----------WA
Tupaia_chinensis
---------------------------CSQPFPNSTNLQPAELSQSLLYRGP-VDPANWF
M_Musculus
---------------------------CTEPFPNNTNLQPLEINQSLLYRGP-VDPANWF
Pan_troglodytes
---------------------------CTEPSPNSTNLLPTEISQSLLYRGP-VDPANWF
Rhinopithecus_roxellana
---------------------------CTEPSPNSTNLLTTEISQSLLYRGP-VDPANWF
                                                         :     .       :

Panicum_hallii
ISSNPPYSS-VRSNEENHSSSRFTSLFSKVQGLIGSSSSSSLDPGNTYQKSEYVDNAIKG
Herrania_umbratica
LSAEDTFTNGFMSNGEDKPSSSFGAVPTRQDRAV--SDSWSFLNPAFSQAPHPVSSKAGG
Mesorhabditis_belari
--------------GFKKNQNMFNELGGLLVSIC----ALTLQSIVIYRQRNFR------
Tupaia_chinensis
--------------GVRKGFPNLGYIQSHLQILL----LLVFEAMVYRRQEHYR------
M_Musculus
--------------GVRKGYPNLGYIQNHLQILL----LLVFEAVVYRRQEHYR------
Pan_troglodytes
--------------GVRKGFPNLGYIQNHLQVLL----LLVFEAIVYRRQEHYR------
Rhinopithecus_roxellana
--------------GVRKGFPNLGYIQNHLQVLL----LLVFEAIVYRRQEHHR------
                                              :    :  :               :
.

Panicum_hallii
SDED--KRYSFAKIWGLSKESHKWDKKRVISLKRERFETQKITFKCYMKFWIENLFKLRG
Herrania_umbratica
SEVSSFRKFSLGYFWGSTKESHKWNKKRILALRKERFETQKALLKLYLKFWMENMFNLYG
Mesorhabditis_belari
------IRTSQPP-----------NLRHRIFPELDPVTYDHSLVNC-FKFFVDFGFYKFG
Tupaia_chinensis
------RQHQLGP-----------PPARSVCVDGTRQRLDQGVLSC-LKYFINFFFYKFG
M_Musculus
------RQHQQAP-----------LPAQAVCADGTRQRLDQDLLSC-LKYFINFFFYKFG
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Pan_troglodytes
------RQHQLAP-----------LPAQAVFASGTRQQLDQDLLGC-LKYFINFFFYKFG
Rhinopithecus_roxellana
------RQHQLAP-----------LPAQAVFASGTRQQLDQDLLGC-LKYFINFFFYKFG
                                      . .                 . :         ::  .
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Panicum_hallii
LEINMVVLLLASFTLLNVVSIFYIMCL-VVCILMNRDLIQKLWPLFVFLFASVLILEYLA
Herrania_umbratica
LEINMIALLLASFALLNAISMLYISLL-AVCVLLNRRIIRKLWPVLVFLFASILILEYFA
Mesorhabditis_belari
LECCLVMIAINAWIRMDMLGAVVIGWL-AVFALSKRTMARWLWPIFLIYLVLLLPIQFS-
Tupaia_chinensis
LEICFLMAVNVIGQRMDFMAILHGCWLGAILTRRRRVAIASLWPNYCLFLTLFLLYQYL-
M_Musculus
LEICFLMAVNVIGQRMNFMVILHGCWLVAILTRRRREAIARLWPNYCLFLTLFLLYQYL-
Pan_troglodytes
LEICFLMAVNVIGQRMNFLVILHGCWLVAILTRRHRQAIARLWPNYCLFLALFLLYQYL-
Rhinopithecus_roxellana
LEICFLMAVNVIGQRMNFMVTLHGCWLVAILTRRHRQAIARLWPNYCLFLALFLLYQYL-
                             **  ::         :: :  .    * .:    .*     ***
: :. .*  ::

Panicum_hallii
LWNNLMPWFHDI-NNIEVHCREC-WK---NSRIFFDYCSKCWLGI----IADDPRMLISY
Herrania_umbratica
IWKNMFPLNQNKPSQAEIHCHDC-WR---SSSSYFQYCRSCWLGL----IIDDPRMLFSY
Mesorhabditis_belari
IYVGLPPSL----------CIDYPWKNLFITNPTKRRNLLIWLGLADYDIQWTSANLLAD
Tupaia_chinensis
LCLGIPPAL----------CLDYPWR--WSRAVPMNSALIKWLYLPDFFRAPNSTNLIIR
M_Musculus
LCLGMPPAL----------CIDYPWR--WSKAIPMNSALIKWLYLPDFFRAPNSTNLISD
Pan_troglodytes
LCLGMPPAL----------CIDYPWR--WSRAVPMNSALIKWLYLPDFFRAPNSTNLISD
Rhinopithecus_roxellana
LCLGMPPAL----------CIDYPWR--WSRAVPMNSALIKWLYLPDFFRSPNSTNLISD
                             :  .: *           * :  *.              ** :
.  *:

Panicum_hallii
YVVFLFSSFKLRSDRFSGFSDSDTYHEMMSQRKNAVVW-----RDLSLETKCFWTFLDYV
Herrania_umbratica
FVVFLLACFKLRADHLSDFSGSSTYRQMMSQRKNSFVW-----RDLSFETKSMWTFLDYL
Mesorhabditis_belari
F--FLLMFVSCQAAVFRVEGPTHPAGDNDSIYKDGVYQFKN--NPRYDFVTTQRSFVDYF
Tupaia_chinensis
A-------------------PSSEWA-----------------RPTQQPRHV-WAGL---
M_Musculus
F--LLLLCASQQWQVFSAE-RTEEWQRMAGINTDHLEPLRGEPNPIPNFIHC-RSYLDML
Pan_troglodytes
F--LLLLCASQQWQVFSAE-RTEEWQRMAGVNSDRLEPLRGEPNPVPNFIHC-RSYLDML
```

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Rhinopithecus_roxellana
F--LLLLCASQQWQVFSAE-RTEEWQCMAGVNTDRLEPLRGEPNPVPNFIHC-RSYLDML
                                                    :                       .
.: :

Panicum_hallii
RLYAYCHLLDIVLALIAITGTLEYDVLHLGYL--GFALVFFRMRLEILKKKN---KIFKY
Herrania_umbratica
RLYCYCHLLDLVLVLILITGTLEYDILHLGYL--AFALVFFRMRLEILKKKN---KLFKF
Mesorhabditis_belari
KKFVFLYAQWMTLVTVMVAGLGGTSLFALGYLCLAFWILWSGNALYVMKNHNRTIKKFMI
Tupaia_chinensis
---------WVCADVRLGQHVCR---------------------------GVSSAYWD-
M_Musculus
KVAVFRYLFWLVLVVVFVAGATRISIFGLGYLLACFYLLLFGTTL-LQKDTRAQLVLWDC
Pan_troglodytes
KVAVFRYLFWLVLVVVFVTGATRISIFGLGYLLACFYLLLFGTAL-LQRDTRARLMLWDC
Rhinopithecus_roxellana
KVAVFRYLFWLVLVVVFVTGATRISIFGLGYLLACFYLLLFGTAL-LQRDTRARLVLWDC
                                         :
:

Panicum_hallii
LRMYNFVLIVLSLAYQSP---YVGQFS--SGT--CDQIDYLY----------EIIGFY--
Herrania_umbratica
LRIYNFAVIVLSLAYQSP---FVGEFS--SGK--CKTVNYIY----------EVIGFY--
Mesorhabditis_belari
VLGYTVFSMFCKVSLQVVGCVFLGSLHGIPGKWGCILRQIFSIACVNNLAMEDIANLF--
Tupaia_chinensis
--------------MPLLSCVFVEQMQ--SSF--CWVIQLFSLVC-------TVKGYYDP
M_Musculus
LILYNVTVIISKNMLSLLSCVFVEQMQ--SNF--CWVIQLFSLVC-------TVKGYYDP
Pan_troglodytes
LILYNVTVVISKNMLSLLACVFVEQMQ--TGF--CWVIQLFSLVC-------TVKGYYDP
Rhinopithecus_roxellana
LILYNVTVIISKNMLSLLACVFVEQMQ--TGF--CWVIQLFSLVC-------TVKGYYDP
                                                    :: .:   ..   *   : :
: . :

Panicum_hallii
------KYDYGFKITSRSAFVEIVIFLLVSVQSYIFSSGEFDYVSRYLEAEQI-------
Herrania_umbratica
------KYDYGFRITARSAIVEIIIFMLVSLQSYMFSSQESDYVSRYLEAEQI-------
Mesorhabditis_belari
PESTEFDKDCPVSAAETRIGFDTLALAFLVLQLRIFHSWYFQHAMTDFRAEIVLANRGAV
Tupaia_chinensis
QATTGRDLDCLLPVEEAGVLWDSICFSFLLLQRRVFLSHYFLHVRADLQATALQASRGFA
M_Musculus
KEMMTRDRDCLLPVEEAGIIWDSICFFFLLLQRRIFLSHYFLHVSADLKATALQASRGFA
Pan_troglodytes
KEMMDRDQDCLLPVEEAGIIWDSVCFFFLLLQRRVFLSHYYLHVRADLQATALLASRGFA
Rhinopithecus_roxellana
KEMMGRDQDCLLPVEEAGIIWDSICFFFLLLQRRVFLSYYYLHVRADLQATALLASRGFA
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: *  :

Panicum_hallii
---GAMVREQEKKALKKTEQLQHLRRSEEHKRQRNMQVERMKSEMYNLQSQLNRMNSFTP
Herrania_umbratica
---GAIVREQEKKAAWKTAQLQQIRESEEKKRQRNFQVEKMKSEMLNLQIQLHSMNSVAT
Mesorhabditis_belari
LANQLIEKEMKEQTQQQNEKFNDIKRRTDDIRAQYQKQQA-------LERTFEPKTYGQA
Tupaia_chinensis
LYNAASLKRIDSRRKAEEKSLAQLKRQMERIRTKQQKHRQGGPGRGRPQDPLGPEEPGQE
M_Musculus
LYNAANLKSINFHRQIEEKSLAQLKRQMKRIRAKQEKYRQSQASRGQLQSK-DPQDPSQE
Pan_troglodytes
LYNAANLKSIDFHRRIEEKSLAQLKRQMERIRAKQEKHRQGRVDRSRPQDTLGPKDPGLE
Rhinopithecus_roxellana
LYNAANLKSIDFHRRTEEKSLAQLKRQMERIRAKQEKHRQGRADRGRPQDTLGPKDPGPE
                                     .  . .   :  .:  :.   .  * .  :
:

Panicum_hallii
INDTS-YNEGLRRR-------------RNTKLY---SDTNTPDVGIETGSPTKQDKIGS-
Herrania_umbratica
LSDVSPDDEGLRRR-------------RSASVT-----------SNRDVVPPDKEEGTL-
Mesorhabditis_belari
K--------------------------RAGDYYMFEKDPENLVPPIESFVPEVDPEGSEF
Tupaia_chinensis
PGPGSPGGSSPPRRQWWRPWLDHATVIHSGDYFLFESDSEEDEEEEEEALPEDPRPSGQ-
M_Musculus
PGPDSPGGSSPPRRQWWRPWLDHATVIHSGDYFLFESDSE----EEEEALPEDPRPAAQ-
Pan_troglodytes
PGPDSPGGSSPPRRQWWRPWLDHATVIHSGDYFLFESDSE----EEEEAVPEDLRPSAQ-
Rhinopithecus_roxellana
PGPDSPGGSSPPRRQWWRPWLDHATVIHSGDYFLFESDSE----EEEEAAPEDPRPSAQ-
                                                          .  .
*

Panicum_hallii
---TESAQSFEFSVTDTQKNIR---------------NLMFQGSSDTMRSPIRGRSDEFV
Herrania_umbratica
------------W--KQEQLIR-------------EEVFPLEA----QADAARMKGESPE
Mesorhabditis_belari
RRLDPTQLVYTAVNKDMNLSASMRQVEKAELIKDDEERRMIAA-VTPDTPPTERVVDEPI
Tupaia_chinensis
---SAFQMAYQAWVTSAQTVLR----RQERGRQ--AEQLPVGGGPSQEVEPAEGPGDEPA
M_Musculus
---SAFQMAYQAWVTNAQTVLR---QRRERARQERAEQLASGGDLNPDVEPVDVPEDEMA
Pan_troglodytes
---SAFQLAYQAWVINAQTVLRRRQQEQEQARQEQAGQLPTGGGPSQEVEPAEGPEEAAA
Rhinopithecus_roxellana
---SAFQMAYQAWVTNAQTVLR--RQQQEQARQEHAGQLPTGG-PSQEVEPAEGPEEAVA
                                             . :                        .
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Panicum_hallii
LTDNIGNSLGSTPEITELEESDEKVNYNLSKWEKAMGQPK---ENPLKSAVQLIGDGVSQ
Herrania_umbratica
VVQSPKQSMVYAP--CEITEIEHDVDSAFCDTEKRKSQAK---ENPLISAVHLLGDGVSQ
Mesorhabditis_belari
AEES-DSFFIATL--KFVFKFAYNVFDWTAVFLNRRSREH-------RYVAYVLGREKQR
Tupaia_chinensis
GRSHVVQRVLSMA--QFLWVLGQALVDGLTRWLQAFTRHHGTMSDVLRAERYLLTQQLLQ
M_Musculus
GRSHMMQRVLSTM--QFLWVLGQATVDGLTRWLRAFTKHHRTMSDVLCAERYLLTQELLR
Pan_troglodytes
GRSHVVQRVLSTA--QFLWMLGQALVDGLTRWLQEFTRHHGTMSDVLRAERYLLTQELLQ
Rhinopithecus_roxellana
GRSHMVQRVLSTA--QFLWVLGQALVDGLTRWLQEFTRHHSTMSDVLRAERYVLTQELLQ
                                .   . .       :               .   . :
::     .

Panicum_hallii
VQSFGNQAVTNI-------------------------------VSFLNIDPDEPHS---N
Herrania_umbratica
VQSIGNQAVNNL-------------------------------VNFLNIAPEDSDM---N
Mesorhabditis_belari
LKNEYGEIL---------------------------------------CDSTKPMA----
Tupaia_chinensis
GGEVDRGVLDQLYASEAGDTQAGTSEARDAPSTVSSGLGAEEPLSSLTEDTSSPLSTGYN
M_Musculus
VGEVRRGVLDQLYVGEDEATLSGPMETRDGPSTASSGLGAEEPLSSMTDDTSSPLSTGYN
Pan_troglodytes
GGEVHRGVLDQLYTSQAEATLPGPTEAPNAPSTVSSGLGAEEPLSSMTDDTGSPLSTGYH
Rhinopithecus_roxellana
GGEVHRGILDQLYTSEAEATLPGPTEAPDAPSTVSSGLGAEEPLSSMTDDTGSPLSTGYH
                                .     :
. ..

Panicum_hallii
EHPA-EGDIYD--VVESQVETR--------DGQLLRTHS------------VSDTSGTKV
Herrania_umbratica
EHSSVEDEAYDEMESQKMQNMC-----------LNRSSS-------------LQSDKSFD
Mesorhabditis_belari
-------------DLQALVPLQ--------QLQVVRTETDI---------EKLESEALGD
Tupaia_chinensis
TRSSSEEIATDPGDPEGEPSLRSSRELPVSARTRTRTASELLLHRRLRIPELEEAARFAE
M_Musculus
TRSGSEEIVTDAGDLQAGTSLHGSQELLANARTRMRTASELLLDRRLHIPELEEAERFEA
Pan_troglodytes
TRSGSEEAVTDPGEREAGASLY---------QGRMRTASELLLDRRLRIPELEEAELFAE
Rhinopithecus_roxellana
TRSGSEEAVTDPGEREAGASLH---------QGRMRTASELLLDRRLRIPELEEAQLFAE
                                            :                  *: :
::

Panicum_hallii
KSSMPI-GVIFRYIWYQMRSNYDYVCYCCFVLVFLWNFSLLSMVYLGALFLYALCVNYGP
```

```
Herrania_umbratica
ATSLQL-GRIFCHIWSQMRSNNDVVCYCFFVLVFLWNFSLLSMVYLAALFLYALCVNTGP
Mesorhabditis_belari
WQKRSVFARLFNALGDCVNAHTDVICYVLAIMAHAYCAGLITLPLPLLVFFWGSLSNPRP
Tupaia_chinensis
GQGRAL--RLLRGV-----------------------------------VWG--GAAHP
M_Musculus
QQGRTL--RLLRAGYQCVAAHSELLCYFIIILNHMVTASAASLVLPVLVFLWAMLTIPRP
Pan_troglodytes
GQGRAL--RLLRAVYQCVAAHSELFCYFIIILNHMVTASAGSLVLPVLVFLWAMLSIPRP
Rhinopithecus_roxellana
GQGRAL--RLLQAVYQCVAAHSELLCYFIIILNHMVTASAGSLVLPVLVFLWAMLSIPRP
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Panicum_hallii
SYLFWVIILIYTELNILSQYIYQIIIQHCGLNIHLPLLQRLGFPDDKI----KASFVVSI
Herrania_umbratica
TYIFWVIMLIYTEVYILLEYLYQIVIQHCGLSINSDLLHELGFPAHEI----KSSFVVSS
Mesorhabditis_belari
SKMFWIVLIAITESIIIIKFIFQFGFFEFNTEQSQSLNNKNQYQWNKLFGVQRQSYYAFL
Tupaia_chinensis
SGR---------------------GRGGWGGRGRRPVLS---------------------
M_Musculus
SKRFWMTAIVFTEVMVVTKYLFQFGFFPWNSYVVLRRYENKPYFPPRILGLEKTDSYIKY
Pan_troglodytes
SKRFWMTAIVFTEIAVVVKYLFQFGFFPWNSHVVLRRYENKPYFPPRILGLEKTDGYIKY
Rhinopithecus_roxellana
SKRFWMTAIVFTEVSVVVKYLFQFGFFPWNSHVVLRRYENKPYFPPRILGLEKTDGYIKY
                            :                            .

Panicum_hallii
LPLFLVYISTLLQSSITAKDGEWVPVTEFSFLSTRNNIEEKHCMPYSWRERLKSLHLPVM
Herrania_umbratica
LPLFLVYLFTLLQSSISAKDGEWMPSTDFNLHRRSAHYRTEVLARYSWSERVSELLQFVI
Mesorhabditis_belari
DVTLLIFLF--YHRYNLRRLGLW----------KDANINDTF----SGN-----------
Tupaia_chinensis
----------------LLCYGLW--------DHEEDPLSKEP----GGRKELEAEEGP--
M_Musculus
DLVQLMALF--FHRSQLLCYGLW--------DHEEDRYPKDHCR--SSVKDREAKEEPEA
Pan_troglodytes
DLVQLMALF--FHRSQLLCYGLW--------DHEEESPSKEHDK--SGEEEQGAEE----
Rhinopithecus_roxellana
DLVQLMALF--FHRSQLLSYGLW--------DHEEDSPSKEHDQ--SGRKEQGAEEGP--
                                                  * *
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Panicum_hallii
NLIRMITRGLSRYWMSLTHGAESPPYFVQVTMEVNHWPEDGIQPERIESAINRVLTIAHE
Herrania_umbratica
NMVKLVIRSFCWYWKSLIQGAETPPYFVQVSMDVHLWPEDGIQPERVESGINQLLRVVHD
Mesorhabditis_belari
------------------NTASIDPLELQMTMENTESPEQ--PQEE--------------
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Tupaia_chinensis
---------PGEPGTA--AGATNDHVQLEVSVGPRDRTLE--LPEETEPCDMRPTSLRFR
M_Musculus
KLESQSETGTGHPKEPVLAGTPRDHIQGKGSIRSKDVIQD--PPEDLKPRHTRHISIRFR
Pan_troglodytes
--------GPGVP-----AATTKDHVQVEARVGPTDGTPE--PQVELRPRDTRRISLRFR
Rhinopithecus_roxellana
--------GPGVP-----EATTEDHVHVEARVVPTDGTPE--PQAELRPRDTRRISLRFR
                                                :       :  :       :

Panicum_hallii
ERCQANLSA-SCHCCSKVRIQSIEKSKENSNMALAVLEVVY-AAPAVCQQPGWYNSLTPA
Herrania_umbratica
ERCTEKIPS-HCPFASRVQVQSIERSQENPNMALIVFEVVYASSLTGCTSADWYKSLTPA
Mesorhabditis_belari
--------------------AAEEDAEKKKEM----------------------------
Tupaia_chinensis
RRMGGPEP--PGPAVRGAE---DGAGEEEEEAVAVAGRQKRPRRPRESVRAAGLWRQSFC
M_Musculus
RRKETPGP--KGTAVMETE-HEEGEGKETTERKRPRHTQEK-SKFRERMKAAGRRLQSFC
Pan_troglodytes
RRKKEEGPARKGAAAIEAEDREEEEGEEEKEA--PTGREKRPSRSGGRVRAAGRRLQGFC
Rhinopithecus_roxellana
RRRKE-GPARRGPEAIEAEDREEEEGKEEKEA--PAGREKRPSRTGRRVRAAGRRLQGFC
                                                        .::  :

Panicum_hallii
ADVEREIHESQKAGLFEEINFPYPIVSVIGGGKREIDLYAYYFGADLAVFFLVAMFYQSV
Herrania_umbratica
ADVAKEILRVQHAGFVEEMGFPYKILSVIGGGKREFDLYAYIFVADLTVFFLVAIFYQSV
Mesorhabditis_belari
--------NPFARFFNQLFRPKFRYIR---------DLYPFFFAFDIICIFIITFGYSSF
Tupaia_chinensis
LSL---VYQPLRRFFHDILHSKHRAAT---------DVYALMFLADVIDFIIIIFGFWAF
M_Musculus
VSLAQSFYQPLQRFFHDILHTKYRAAT---------DVYALMFLADIVDIIIIIFGFWAF
Pan_troglodytes
LSLAQGTYRPLRRFFHDILHTKYRAAT---------DVYALMFLADVVDFIIIIFGFWAF
Rhinopithecus_roxellana
LSLAQGTYRPLQRFFHDILHTKYRAAT---------DVYALMFLADVVDFIIIIFGFWAF
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Panicum_hallii
LKNKSEFLEVYQL-EDQFPKEFVFILMVLFFLIVVDRIIYLWSFATGKVVFYIFNLVLFT
Herrania_umbratica
IKNNSEFLDVYQL-EDQFPKEYVFILMIIFFLIVVDRILYLCSFATGKIIFYLFSLVLFT
Mesorhabditis_belari
GEGGSG-NVVSDIQSNRIPLAFVIMLIVLTCMIVIDRGLYLRKAVYCKLAYQLI-SIIFL
Tupaia_chinensis
GKHSAATDITSSLSDDQVPEAFLVMLLIQFGTMVVDRALYLRKTVLGKLAFQVV-LVLAI
M_Musculus
GKHSAATDIASSLSDDQVPQAFLFMLLVQFGTMVIDRALYLRKTVLGKLAFQVV-LVVAI
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Pan_troglodytes
GKHSAATDITSSLSDDQVPEAFLVMLLIQFSTMVVDRALYLRKTVLGKLAFQVV-LVLAI
Rhinopithecus_roxellana
GKHSAATDITSSLSDDQVPEAFLVMLLIQFSTMVVDRALYLRKTVLGKLAFQVA-LVLAI
                              :  :    . .: .:..*  ::.:*::    :*:** :** . .
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Panicum_hallii
YSVTEYAWGMELAHRDVGGLVLRAIYLTKSISLALQALQIRYDIPNKSNLYRQFLTSKVT
Herrania_umbratica
YSITEYAWQIKSSSQNAGRLALRAIFLAKAVSLALQAVQIRHGIPNKSTLYRQFLTSEVS
Mesorhabditis_belari
HIWIFLVLPSITKREASSNRMALLLYFVKCMYFLVSAWQIRNGYPAL--CVGNLLTHAYG
Tupaia_chinensis
HLWMFFVLPAVTERMFRQNTVAQLWYFVKCVYFALSAYQIRWGYPTR--ILGNFLTKKYN
M_Musculus
HIWMFFILPAVTERMFSQNAVAQLWYFVKCIYFALSAYQIRCGYPTR--ILGNFLTKKYN
Pan_troglodytes
HLWMFFILPAVTERMFNQNVVAQLWYFVKCIYFALSAYQIRCGYPTR--ILGNFLTKKYN
Rhinopithecus_roxellana
HLWMFFILPAVTERMFSQNAVAQLWYFVKCIYFALSAYQIRCGYPTR--ILGNFLTKKYN
                            :                        ::.*.: : :.* *** . *
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Panicum_hallii
QVNYLGFRLYRALPFLYELRCVLDWSCTTTSLTMYDWLKLEDIYASLFLVKCDAILNRA-
Herrania_umbratica
RINYLGYRLYRALPFLYELRCVLDWSCTTTSLTMYDWLKLEDINASLYLVKCDAVLNRA-
Mesorhabditis_belari
LTNMVFFKIFMAVPFLFELRTAIDWTWTDTSMPLFDFFNMENFYATIYNLKCGRQFEQSY
Tupaia_chinensis
HLNLFLFQGFRLVPFLVELRAVMDWVWTDTTLSLPSWMCVEDVYANTFIIKCSRETEKKY
M_Musculus
HLNLFLFQGFRLVPFLVELRAVMDWVWTDTTLSLSNWMCVEDIYANIFIIKCSRETEKKY
Pan_troglodytes
HLNLFLFQGFRLVPFLVELRAVMDWVWTDTTLSLSSWMCVEDIYANIFIIKCSRETEKRY
Rhinopithecus_roxellana
HLNLFLFQGFRLVPFLVELRAVMDWVWTDTTLSLSSWMCVEDIYANIFIIKCSRETEKKY
                              * . :. :  :*** *** .:**  * *::.: .:: :*:.
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Panicum_hallii
NHRQGEKQSKMTKFCSGICLFFVLICVIWAPMLIYSSGNP-TNIANPIIDVSIQIDIKAL
Herrania_umbratica
KHKQGEKQTKMTKCCNGICLFFILLCVIWAPMLMYSSGNP-TNMANPIKDASFQTDISTG
Mesorhabditis_belari
PAPRGVEKGSLIKYLMGIPMILFIILIIWSPLIAFSLLNR-IGVVLTPNIARMSISV---
Tupaia_chinensis
PQPKGQKKKKVVKYGMGGLIILFLIAIIWFPLLFMSLVRSVVGVVNQPIDVTVTLKL---
M_Musculus
PQPKGQKKKKIVKYGMGGLIILFLIAIIWFPLLFMSLIRSVVGVVNQPIDVTVTLKL---
Pan_troglodytes
PQPKGQKKKKIVKYGMGGLIILFLIAIIWFPLLFMSLVRSVVGVVNQPIDVTVTLKL---
```

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Rhinopithecus_roxellana
PQPKGQKKKKIVKYGMGGLIILFLIAIIWFPLLFMSLVRSVVGVVNQPIDVTVTLKL---
                                .* :: .: *   *  :::.:: :** *::  *  .   .:.
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Panicum_hallii
GGRL-------------------------------------------------------
Herrania_umbratica
GGRL-------------------------------------------------------
Mesorhabditis_belari
DGYP-------------------------------------------------------
Tupaia_chinensis
GGYEPLFTMSAQLQGPWVEGHFQLGALPTPRPASPQPLFTMSAQLQGPWVEGHFQLGALP
M_Musculus
GGYE-------------------------------------------------------
Pan_troglodytes
GGYE-------------------------------------------------------
Rhinopithecus_roxellana
GGYE-------------------------------------------------------
                            .*

Panicum_hallii
--------TLFKTTACEK-------IPWKYLKAYDD----VDPL------------GYLG
Herrania_umbratica
--------TLYQTTLCEK-------LQWDKLNSDVN----LDPL------------NYLD
Mesorhabditis_belari
--------PLYSIEAQGTELKHLTAAELSGLRSGFSVAFVPGPMQDDSIRKSRSAVSFLD
Tupaia_chinensis
MPRPASPQPLFTMSAQQPSIVPFTPQAYEQLSRQFD----SHPL----------AMQFIS
M_Musculus
--------PLFTMSAQQPSIVPFTPQAYEELSQQFD----PYPL----------AMQFIS
Pan_troglodytes
--------PLFTMSAQQPSIIPFTAQAYEELSRQFD----PQPL----------AMQFIS
Rhinopithecus_roxellana
--------PLFTMSAQQPSIIPFTAQAYEELSRQFD----PHPL----------AMQFIS
                                     .*:                . *    .      *:
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Panicum_hallii
SYNVDDIQLICCQPDASTMWLIPPPVQSRFIQSLEREIPFEKMELILNWDFLRARPKGKE
Herrania_umbratica
TYNKNDIQLICCQADASILWLVPDVVQRRFIQSLDWDM---DMGITSTWLLTRERPKGKE
Mesorhabditis_belari
EYTQDDSLKVLFRPESEQSWDISDDSKTAMIAALNDTTT--TMQMQIHISFTRPRQDSSK
Tupaia_chinensis
QYSPEDIVTAQIEGSSGALWRISPPSRAQMQRELESGSA--DITLRLTWNFQRDLAKGGT
M_Musculus
QYSPEDIVTAQIEGSSGALWRISPPSRAQMKQELYNGTA--DITLRFTWNFQRDLAKGGT
Pan_troglodytes
QYSPEDIVTAQIEGSSGALWRISPPSRAQMKRELYNGTA--DITLRFTWNFQRDLAKGGT
Rhinopithecus_roxellana
QYSPEDIVTAQIEGSSGALWRISPPSRAQMKRELYNGTA--DITLRFTWNFQRDLAKGGT
                              *. :*        .:   * :.   .  :   *        : :
: *   ..
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Panicum_hallii
LVRY---ESPIEHCPSVDDVKQ---VLNGTTHSFSIV-DAYPRYFRVTGSGEVRRL----
Herrania_umbratica
VVQYHKTVDSKDLPDRSDVQK----VLNGSTNSFRIY-NLYPRYFRVTGSGEVRPF----
Mesorhabditis_belari
EPVTHTSEFSIAMDNTTKIREQMLKALRDDTNSTVIVSNAIPVYLIVPNEGEVIQASALN
Tupaia_chinensis
-VEYSNDKHTLDLAPNSTARRQLASLLEGASDQSVV------------------------
M_Musculus
-VEYTNEKHTLELAPNSTARRQLAQLLEGRPDQSVVIPHLFPKYIRAPNGPEANPVKQL-
Pan_troglodytes
-VEYANEKHMLALAPNSTARQQLASLLEGTSDQSVVIPNLFPKYIRAPNGPEANPVKQL-
Rhinopithecus_roxellana
-VEYANEKHMLALAPNSTARRQLASLLEGTSDQSVVIPNLFPKYIRAPNGPEANPVKQL-
                                                        * . . .  :

Panicum_hallii
---------EAAIDSVSGELLLNNDT-----------------PPWWSFY-TKPSDLAGC
Herrania_umbratica
---------EQEVSSVSADLVINHAA-----------------FEWWSFHDINSSNVRGC
Mesorhabditis_belari
WVMNQFVKPELKQNSLATLKLTLNAQ------------------NLWTVE-MQRPELNAT
Tupaia_chinensis
------------------------------------------------------------
M_Musculus
-------QPDEEEDYLGVRIQLRREQVGTGASGEQAGTKASDFLEWWVI------ELQDC
Pan_troglodytes
-------QPNEEADYLGVRIQLRREQ-GAGATG---------FLEWWVI------ELQEC
Rhinopithecus_roxellana
-------QPNEEADYLGVRIQLRREQ-GAGATG---------FLEWWVI------ELQEC


Panicum_hallii
QVL------------NGPMAIVV-SEETPQGIIGETLSKFSIWSLYITFVLAVARFIRLQ
Herrania_umbratica
RDL------------TGPMAIIVSEETPPQGILGDTLSKFSIWGLYITFVLAVGRFIRLQ
Mesorhabditis_belari
NLLLPAEKVKYGTGGKEYLQLVSFVDRTFPSFFAKYVQG-GVIAMYIALVIVVGKVLRGL
Tupaia_chinensis
-------------------------------------------MGLYVSIVLVIGKFVRGF
M_Musculus
KAD------------CNLLPMVIFSDKVSPPSLG-FLAGYGIVGLYVSIVLVVGKFVRGF
Pan_troglodytes
RTD------------CNLLPMVIFSDKVSPPSLG-FLAGYGIMGLYVSIVLVIGKFVRGF
Rhinopithecus_roxellana
RTD------------CNLLPMVIFSDKVSPPSLG-FLAGYGIMGLYVSIVLVIGKFVRGF

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Panicum_hallii
CSDLRMRIPYENLPSCDRLLDICEGIYAARAEGELEVEEVLYWTLVNIYRSPHMLLEYTK
Herrania_umbratica
CSDLRMRIPYENLPSCDRLIAICEDIYAARAEGELGVEEVLYWTLVKIYRSPHMLLEYTK
```

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Mesorhabditis_belari
FTNQPLDVIIAEIPNPDHLLKICLDIYLVREAKDFVLEQDLFAKLIFLFRSPETLIKWTR
Tupaia_chinensis
FSEISHSIMFEELPCVDRILKLCQDIFLVRETRELELEEELYAKLIFLYRSPETMIKWTR
M_Musculus
FSEISHSIMFEELPCVDRILKLCQDIFLVRETRELELEEELYAKLIFLYRSPETMIKWTR
Pan_troglodytes
FSEISHSIMFEELPCVDRILKLCQDIFLVRETRELELEEELYAKLIFLYRSPETMIKWTR
Rhinopithecus_roxellana
FSEISHSIMFEELPCVDRILKLCQDIFLVRETRELELEEELYAKLIFLYRSPETMIKWTR
                              ::    :   ::*  *.:: :* .*: .*   :: :*: *:
.*: ::***  ::::*.

Panicum_hallii               PD----
Herrania_umbratica           PD----
Mesorhabditis_belari         YKFKDD
Tupaia_chinensis             EK---E
M_Musculus                   ER---E
Pan_troglodytes              EK---E
Rhinopithecus_roxellana      EK---E
```
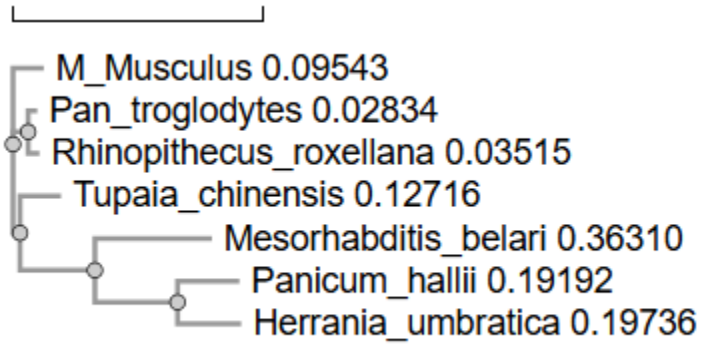
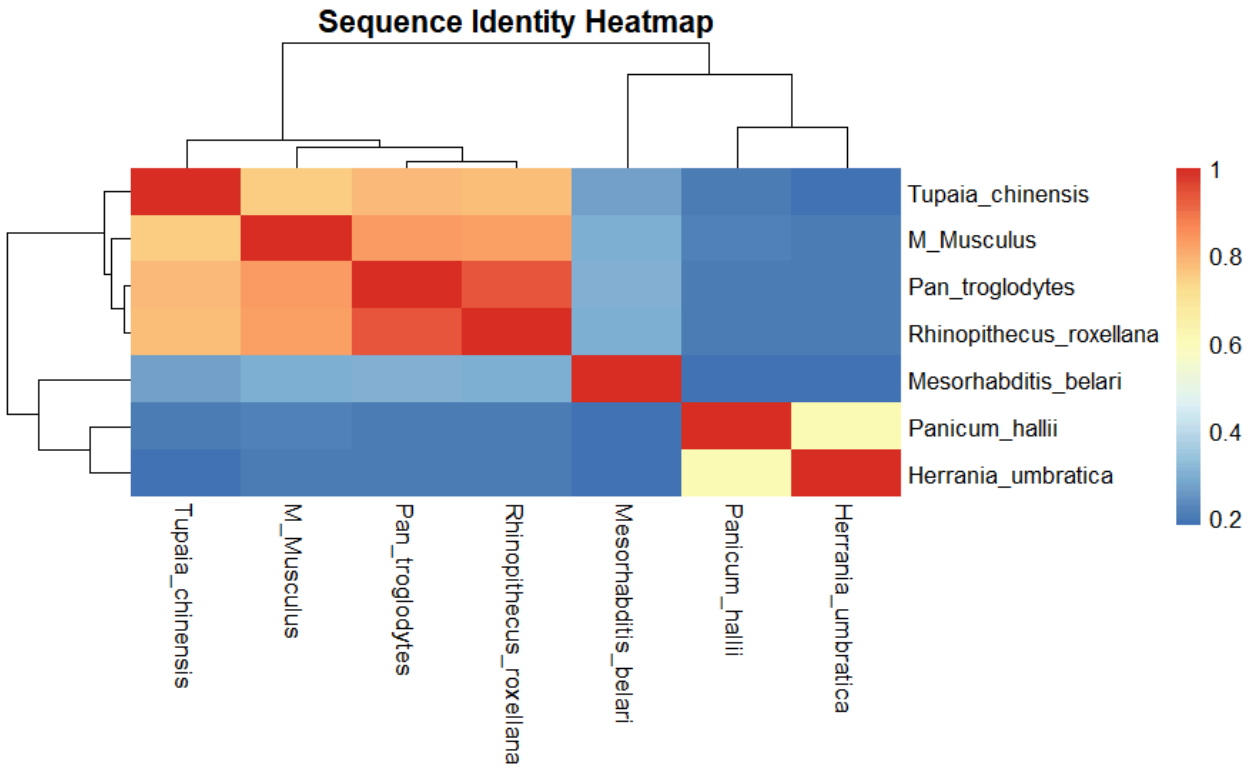

6. Alignment Visualization

Import sequences into EBI Simple Phylogeny, align with MUSCLE, and create a neighbor-joining tree:



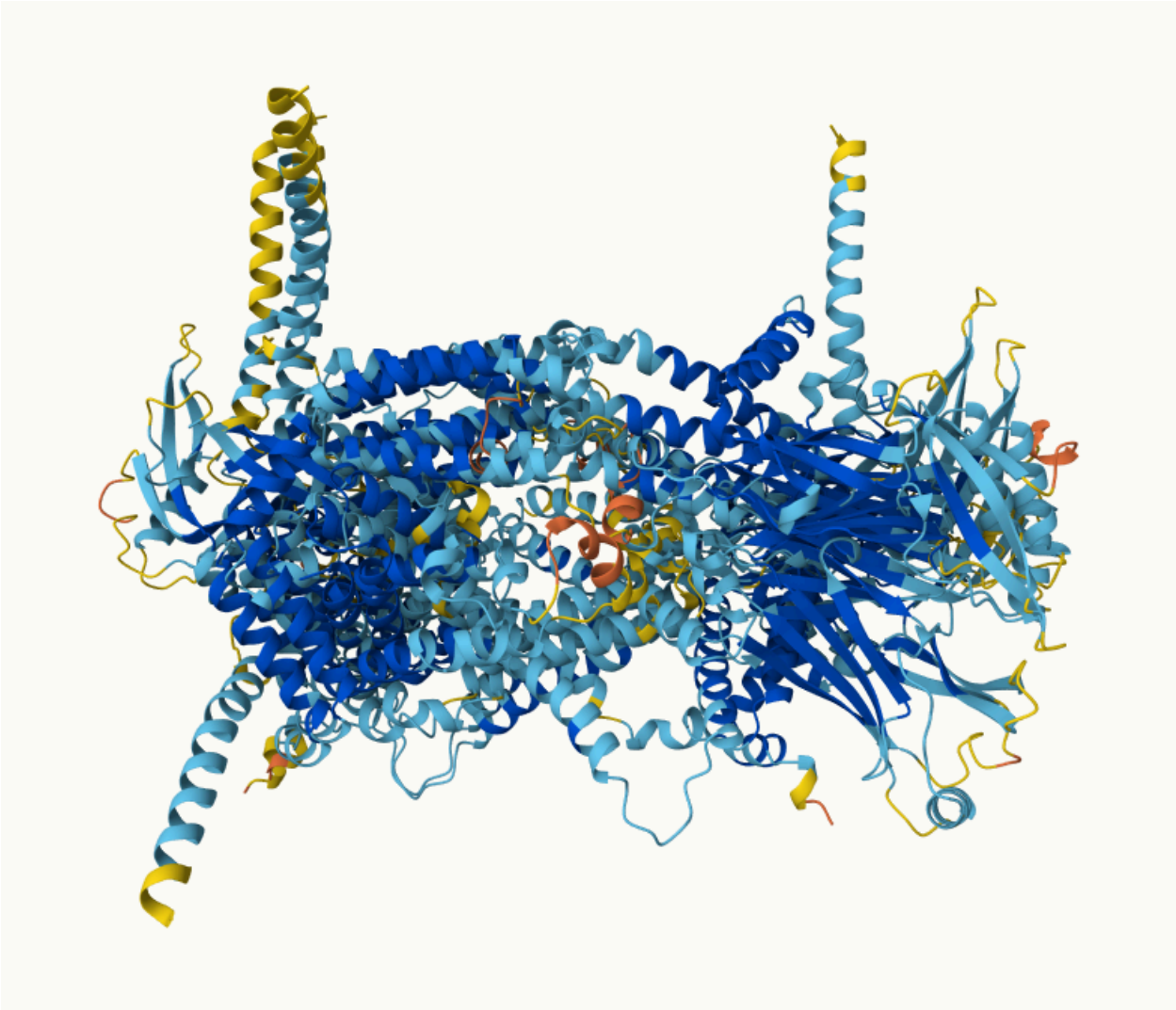
7. Heat Map



8. Related Protein Information

ID	Technique	Resolution	Source	E-value	Identity
8IXN	EM	4.07	Mus musculus	0	99.804
5Z10	EM	3.97	Mus musculus	0	99.843
6LQI	EM	4.50	Mus musculus	0	98.901

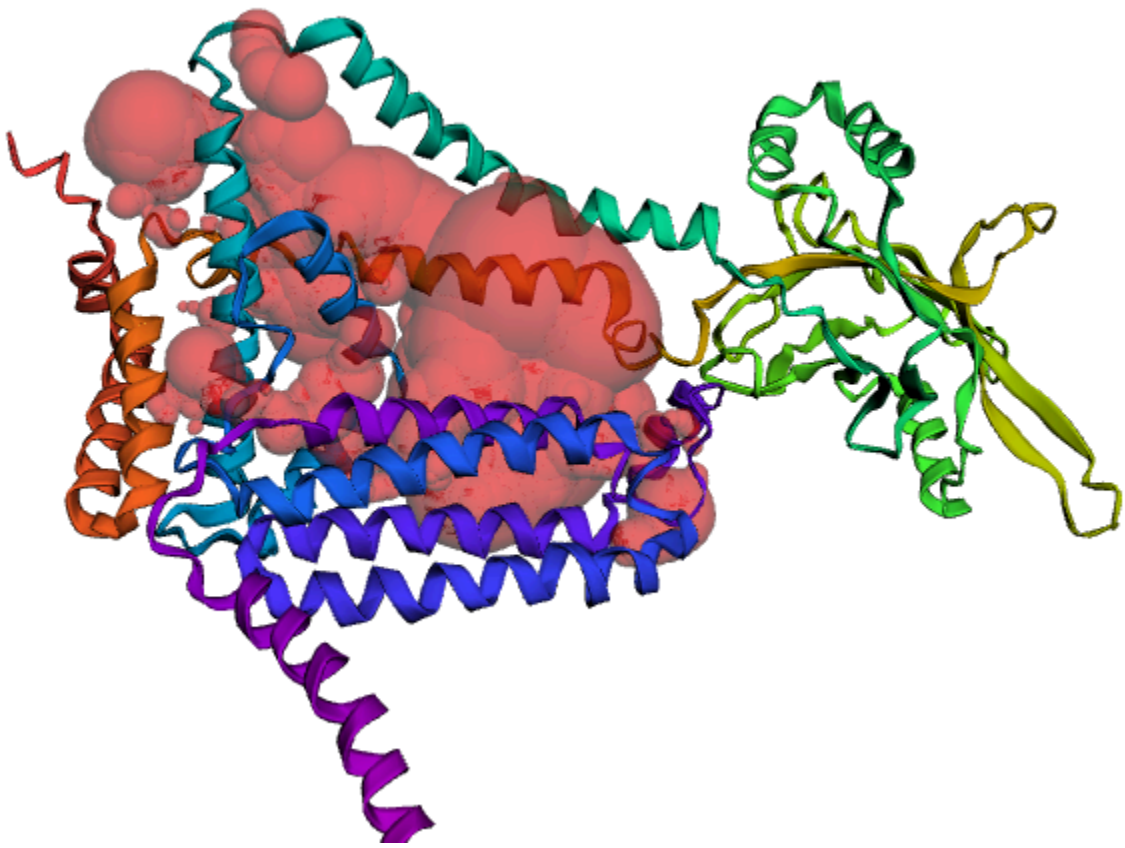
9. Structural Analysis



The pLDDT showed a high confidence landscape since the majority of the structures are in blue and cyan colors, indicating the protein is a well folded domain. There are many alpha helices and loops in the high confidence areas showing the stability of the domain. The high confidence structures form a packed bundle, which can be essential for the function of the PIEZO 1 protein: the high confidence on each side may give the protein the ability to stay on the cell membrane, and the low confidence region in the middle may give the protein pressure censoring ability since it changes shape when sensing pressure.

10. Conclustion

(i)



Pocket ID 1 has the largest negative value. It has an area of 3045.726 and volume of 6580.764.

(ii)

The target CHEMBEL search has matching results with low E-value and high scores in bits. Many of the top matches are from homo sapiens and mus musculus, and they are all signal proteins.

(iii)

Overall, PIEZO 1 has high druggability because it has a generally stable structure and deep pockets, which helps small molecule binding. Although the CHEMBEL search results are mainly

signal proteins, PIEZO 1's wide spread through bacteria and mammals enables it to be produced. Moreover, the PIEZO 1 protein has similar binding sites across different species, which makes research more straightforward. Lastly, PIEZO 1 protein, as an ion channel, has many potential implications for cardiac conduction disorder and even sheds light on myocardial infarction treatment.